



PARTITIONED REGRESSION

Fundamentals of
PROGRAM EVALUATION

JESSE LECY

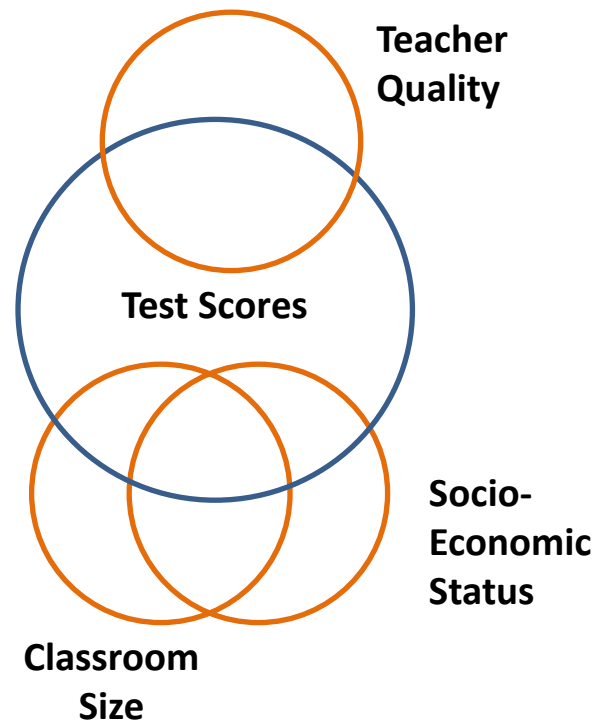
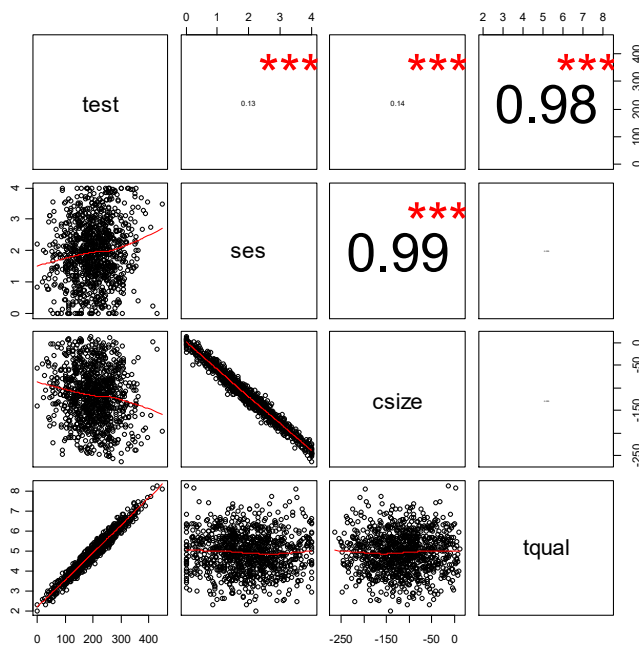
EXAMPLE: CLASSROOM SIZE

	Model 1	Model 2	Model 3	Model 4	Model 5
Constant	141.675*** (4.871)	-179.058*** (1.609)	141.479*** (4.601)	141.553*** (4.596)	-179.725*** (1.575)
Class Size	-0.433*** (0.021)	-0.468*** (0.003)		-0.377 (0.207)	-0.267*** (0.029)
Quality of Instruction		62.125*** (0.285)			62.169*** (0.278)
Socio-Economic Status			43.420*** (2.076)	5.649 (20.853)	20.344*** (2.918)
R-squared	0.307	0.986	0.305	0.307	0.986
N	1000	1000	1000	1000	1000

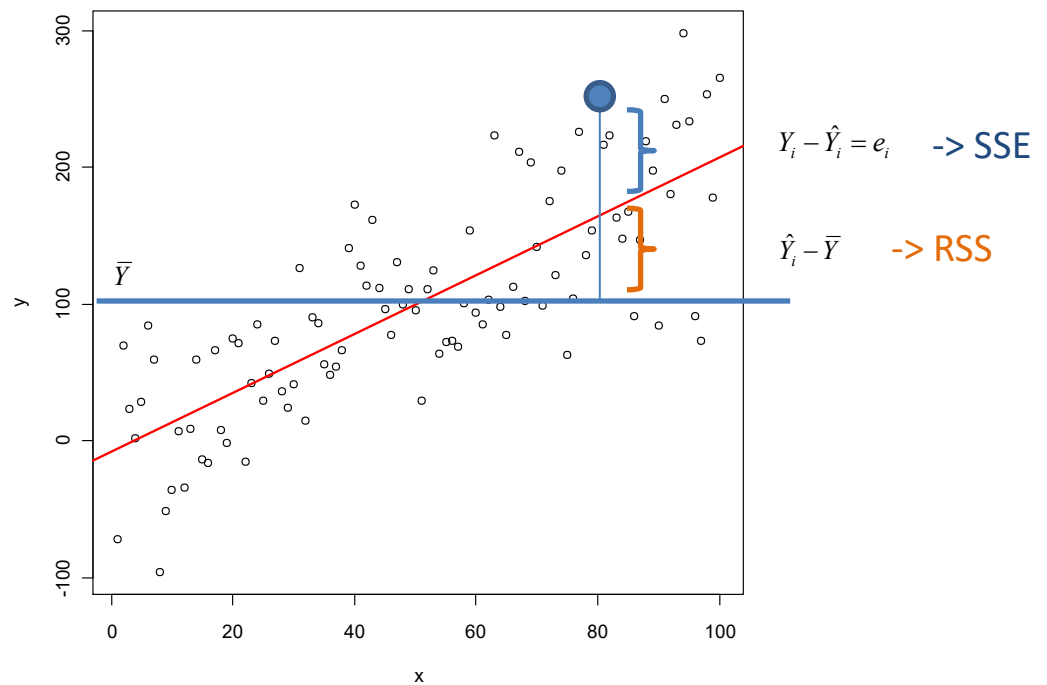
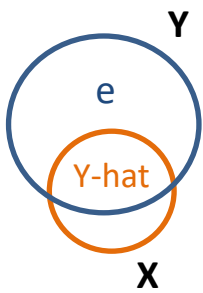
Why are slopes and standard errors changing when we add “control” variables?



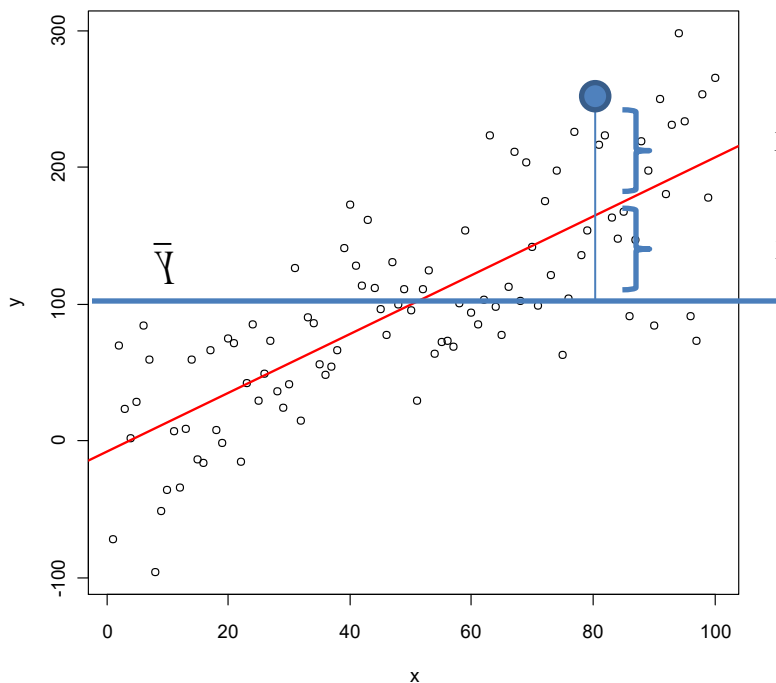
EXAMPLE: CLASSROOM SIZE



PARTITIONING THE VARIANCE OF Y



STANDARD ERROR IN REGRESSION



$$SSE = \sum e_i^2$$

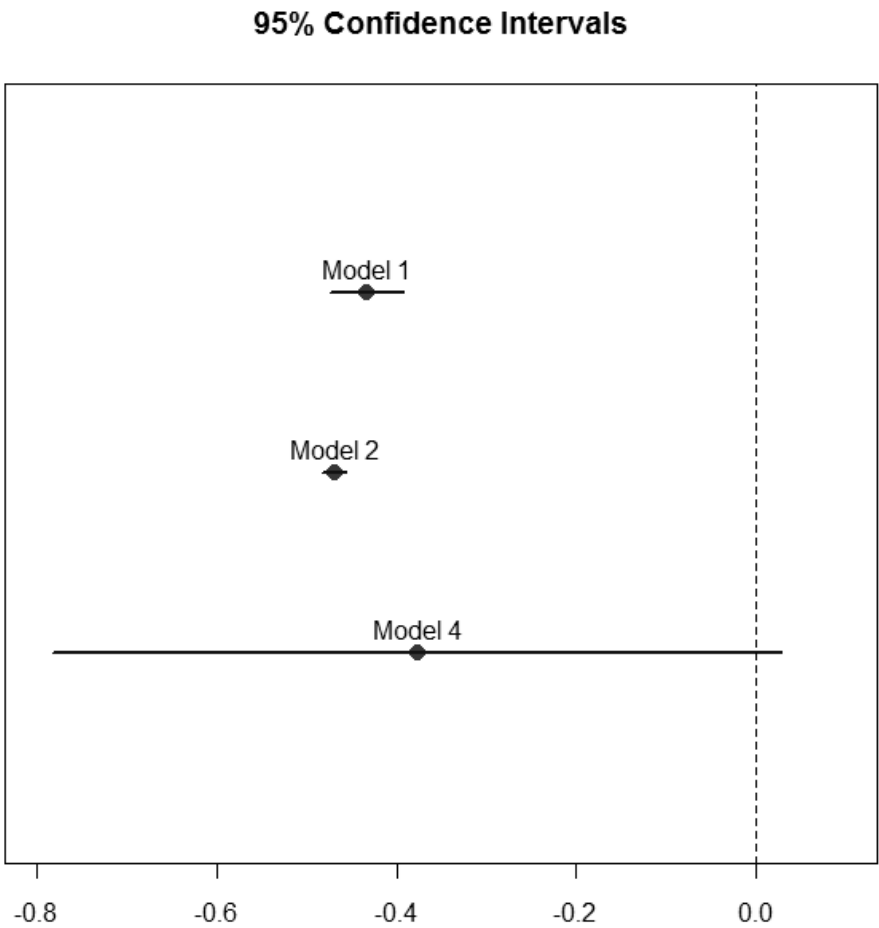
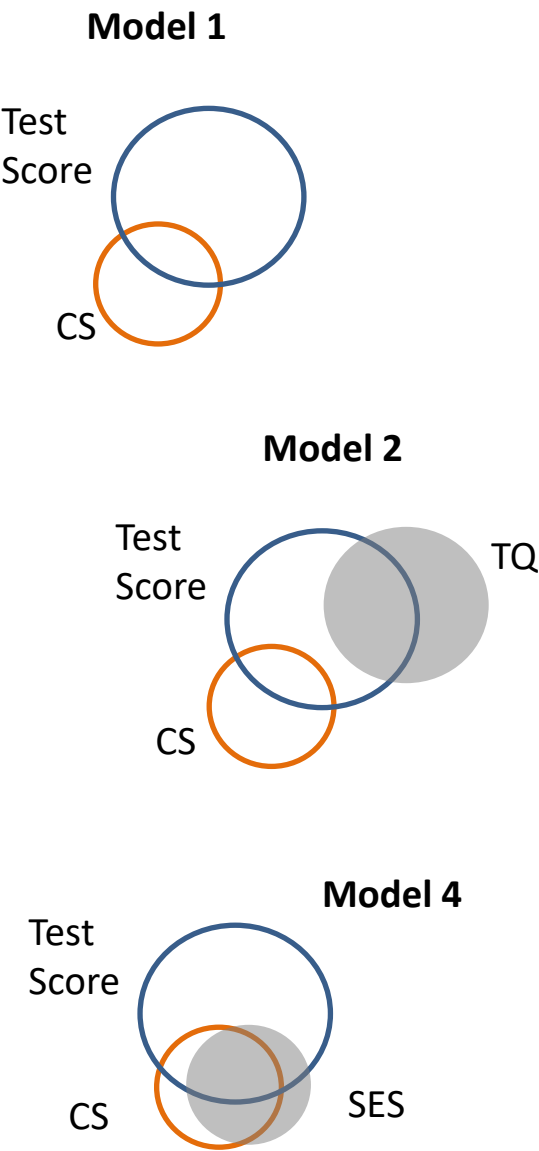
Sum of Squared Error Terms

$$\hat{\sigma}_\varepsilon^2 = \frac{SSE}{n-2}$$

Variance of the residual

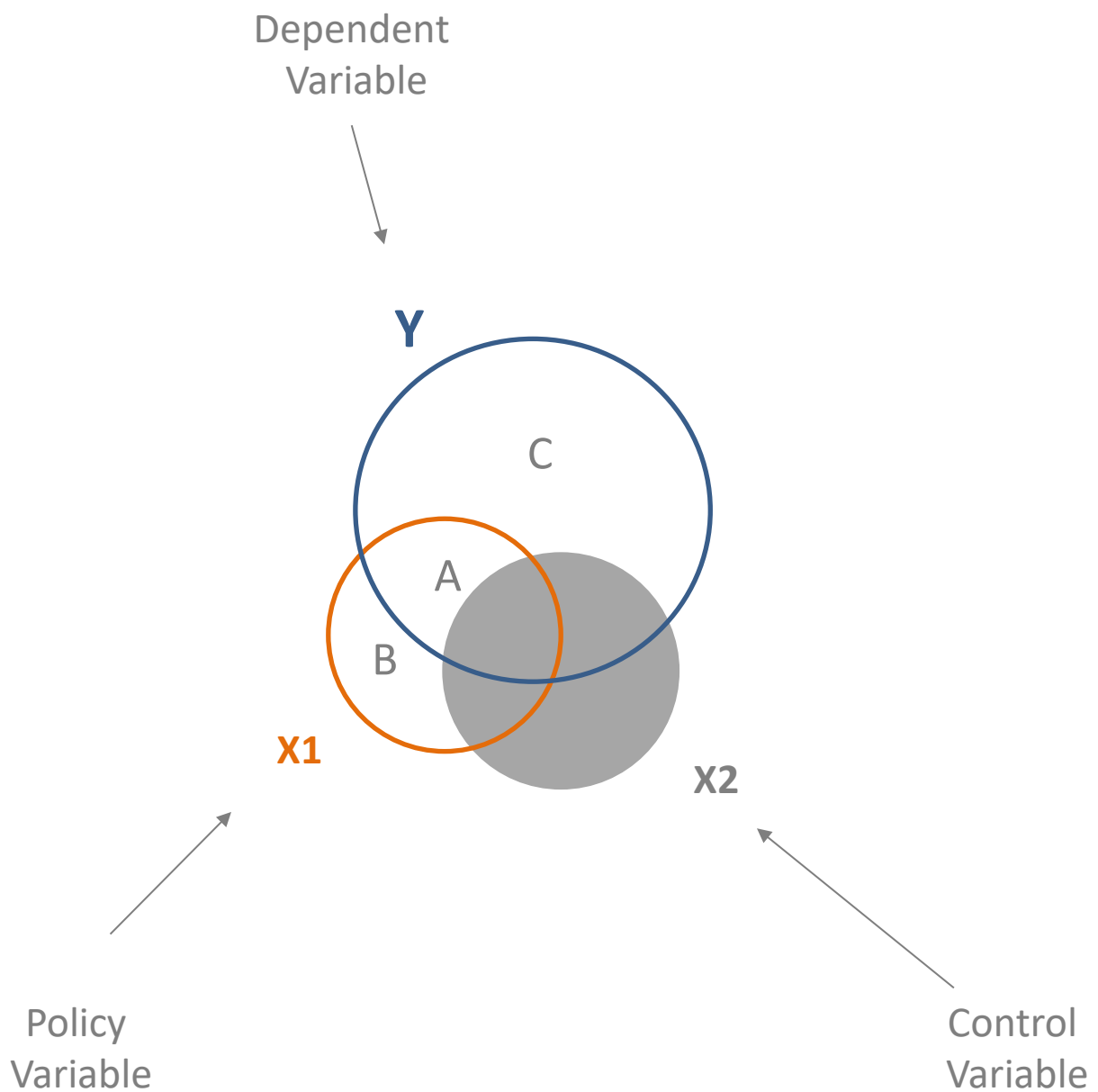
$$SE_{b_1} = \sqrt{\frac{\hat{\sigma}_\varepsilon^2}{\sum (x_i - \bar{x})^2}}$$

Standard error of the slope



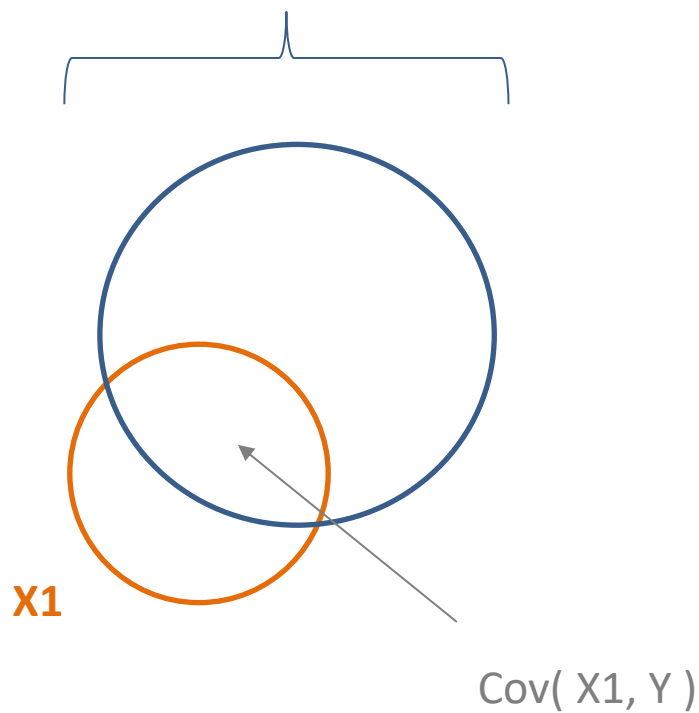
PARTITIONED REGRESSION

BALLENTINE VENN DIAGRAM

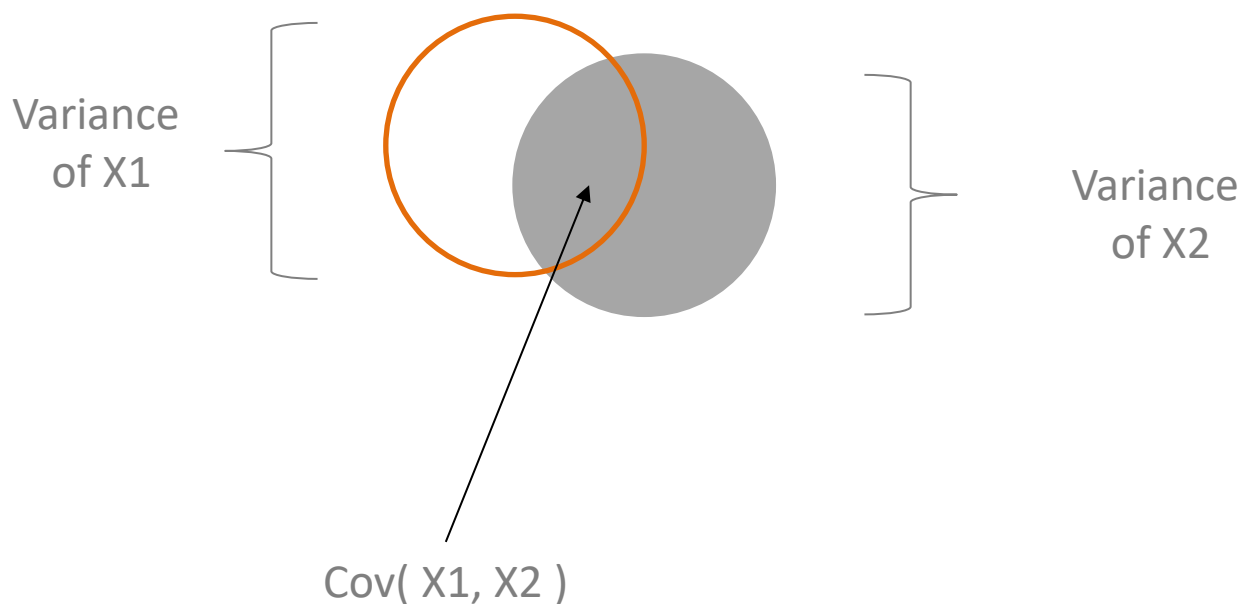


BALLEN'TINE VENN DIAGRAM

Variance of Y



BALLENTINE VENN DIAGRAM

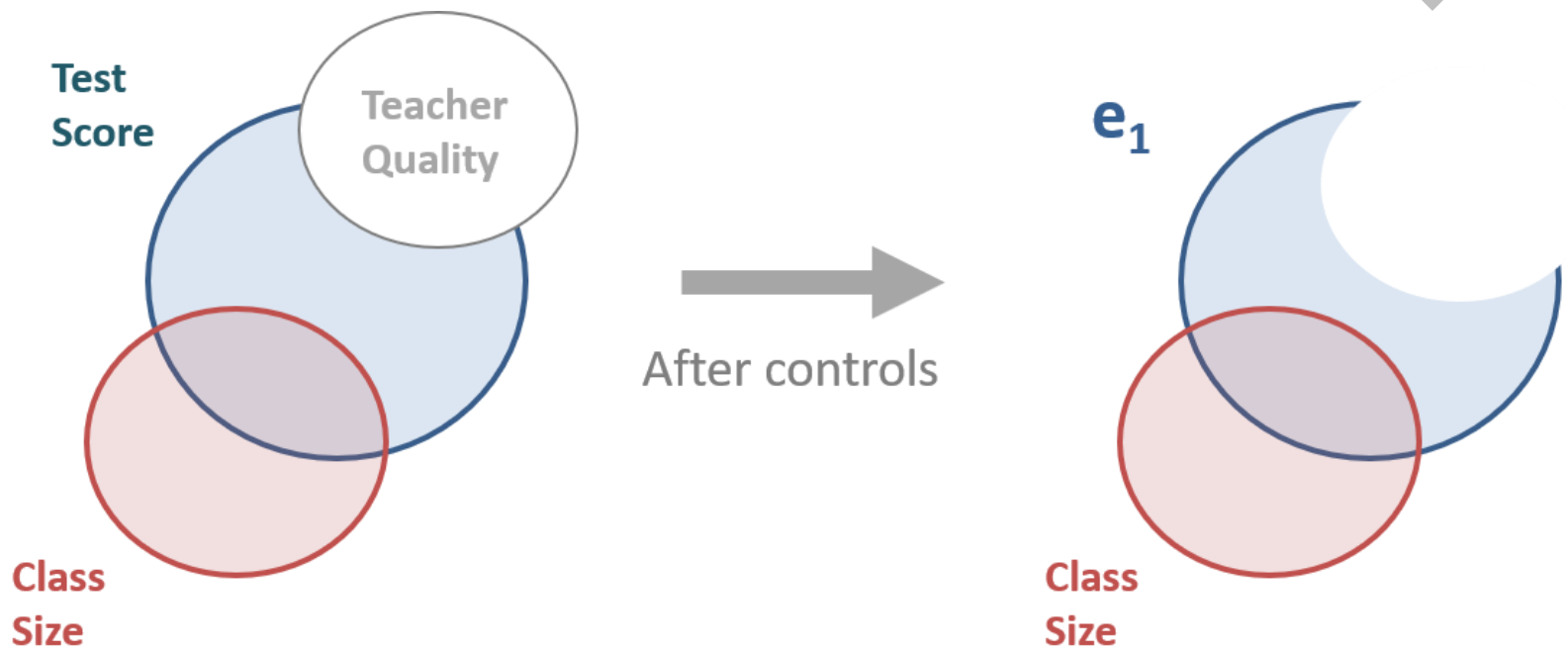




PARTITIONED REGRESSION

Fundamentals of
PROGRAM EVALUATION

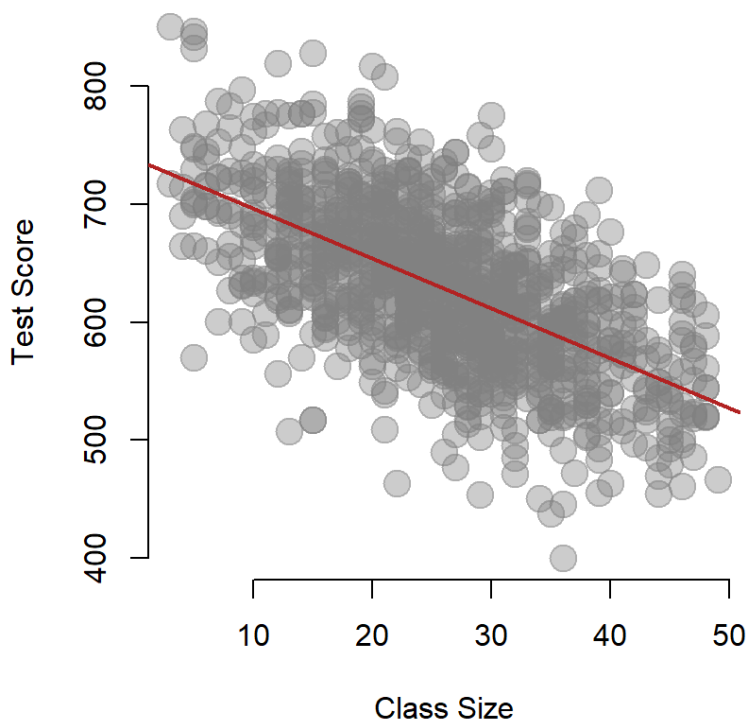
JESSE LECY



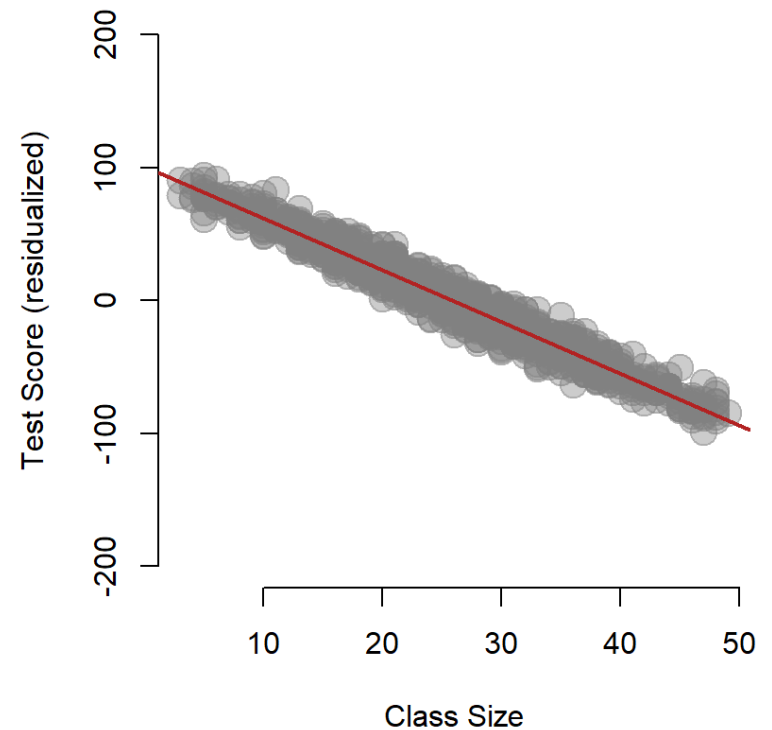
$$test = b_0 + b_1 \cdot tqual + e_1$$

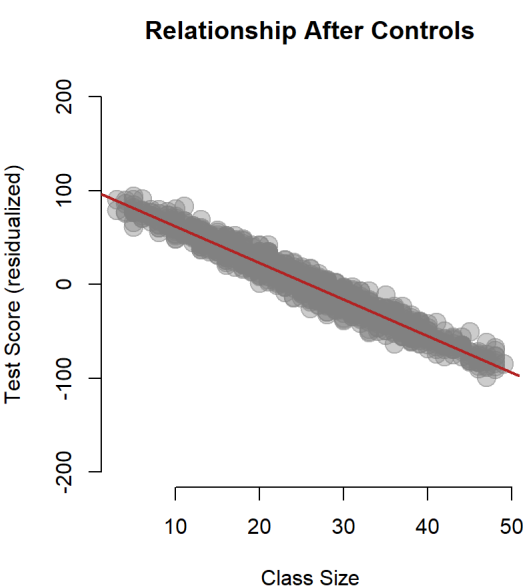
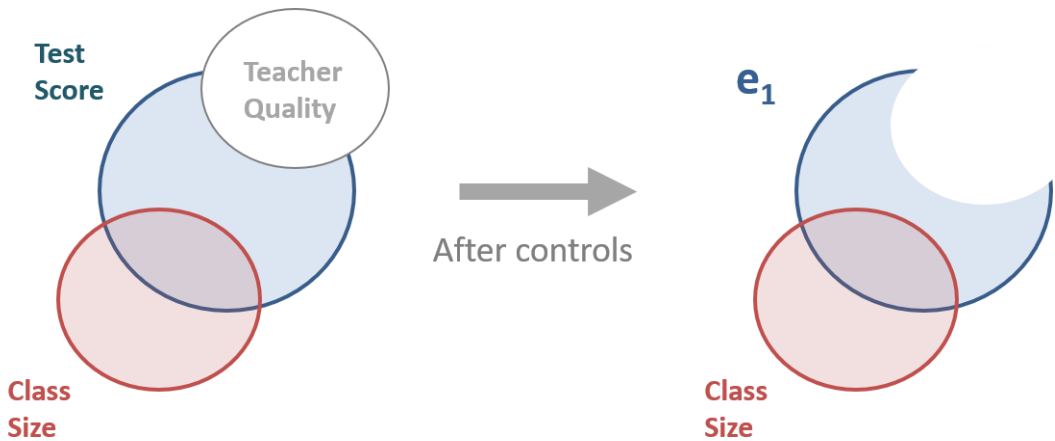
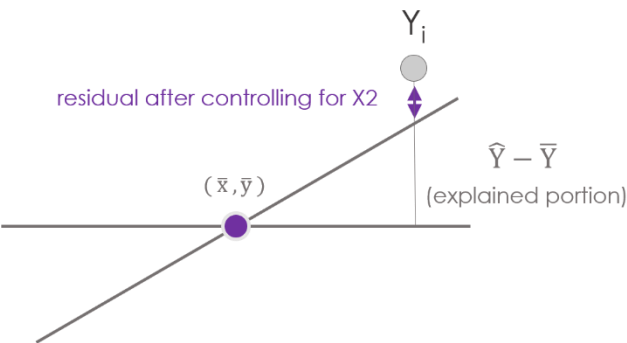
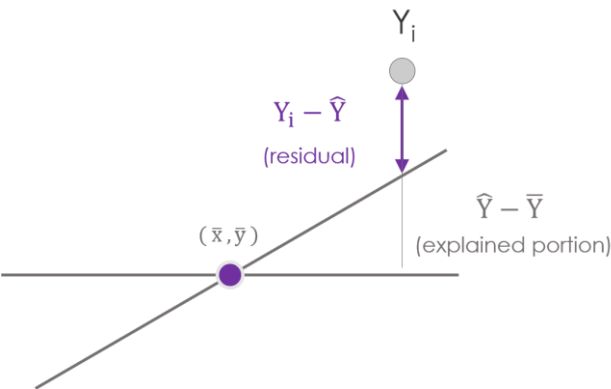
$$e_1 = b_0 + b_1 \cdot csize + e_{e_1}$$

Relationship Between
Class Size and Test Scores



Relationship After Controls





Test
ScoreClass
Size

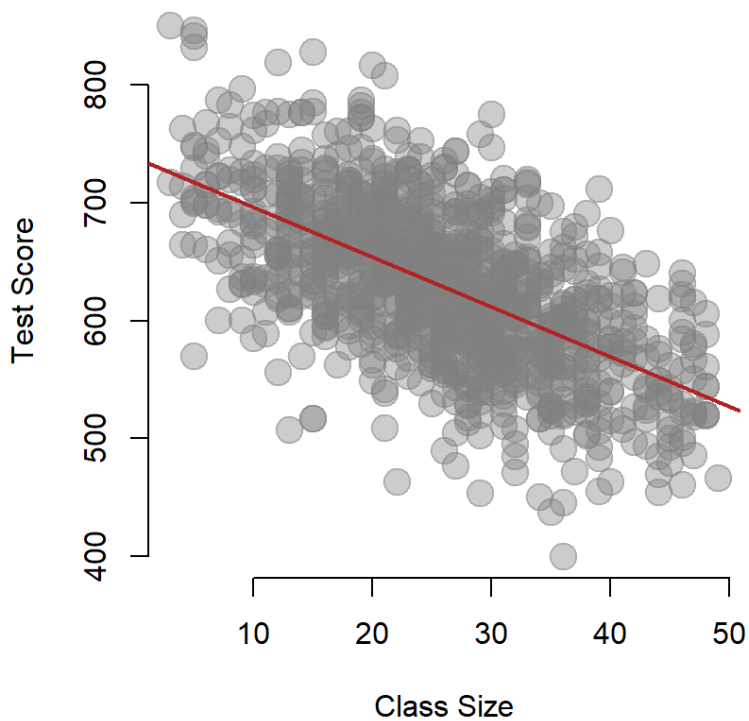
SES

After controls

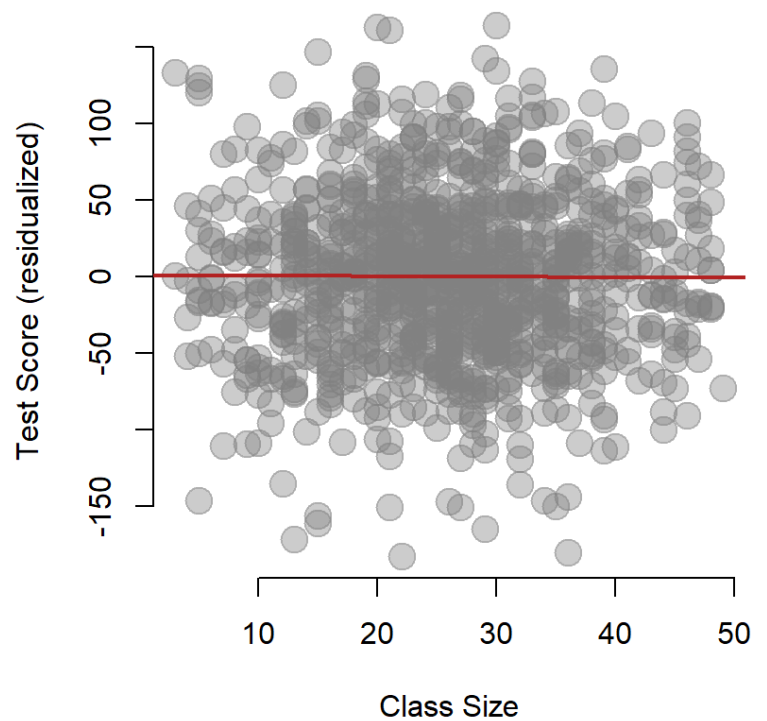
 e_2 Class
Size

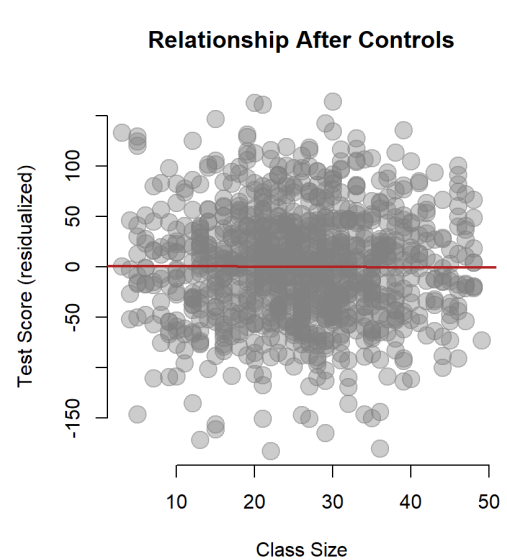
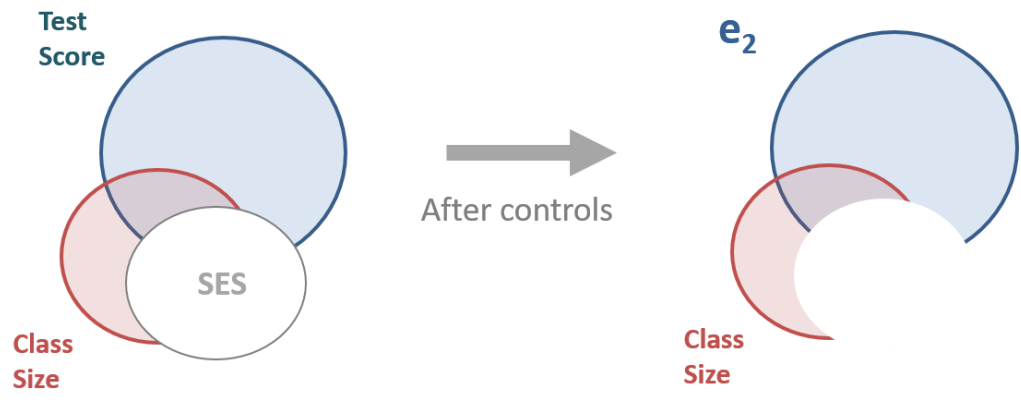
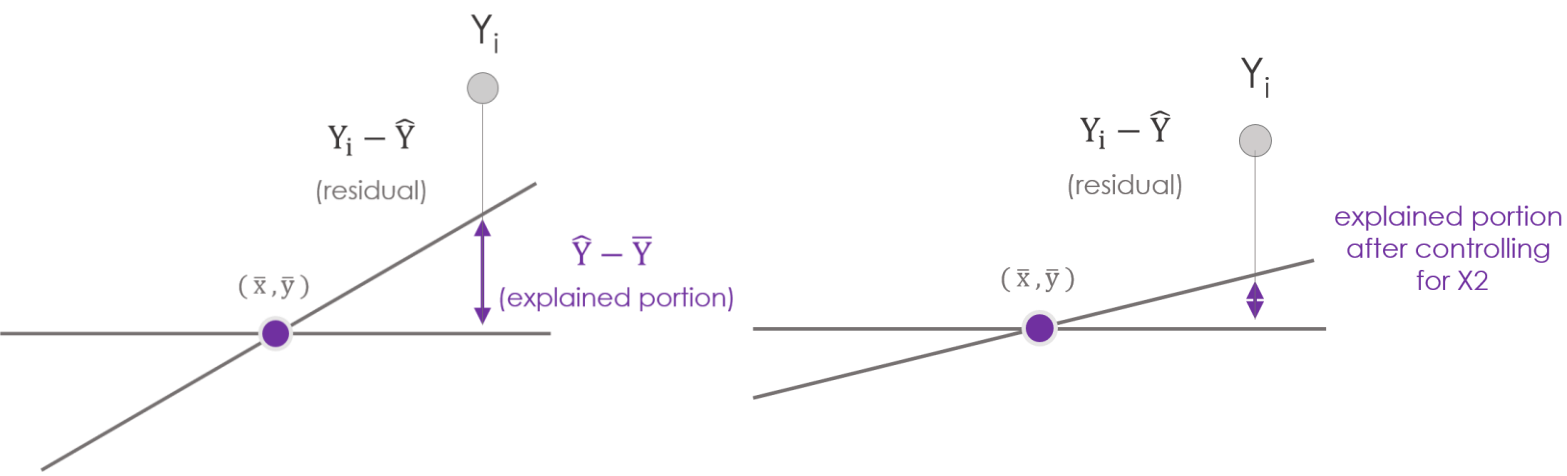
$$test = b_0 + b_1 \cdot ses + e_2$$

$$e_2 = b_0 + b_1 \cdot csize + e_{e_2}$$

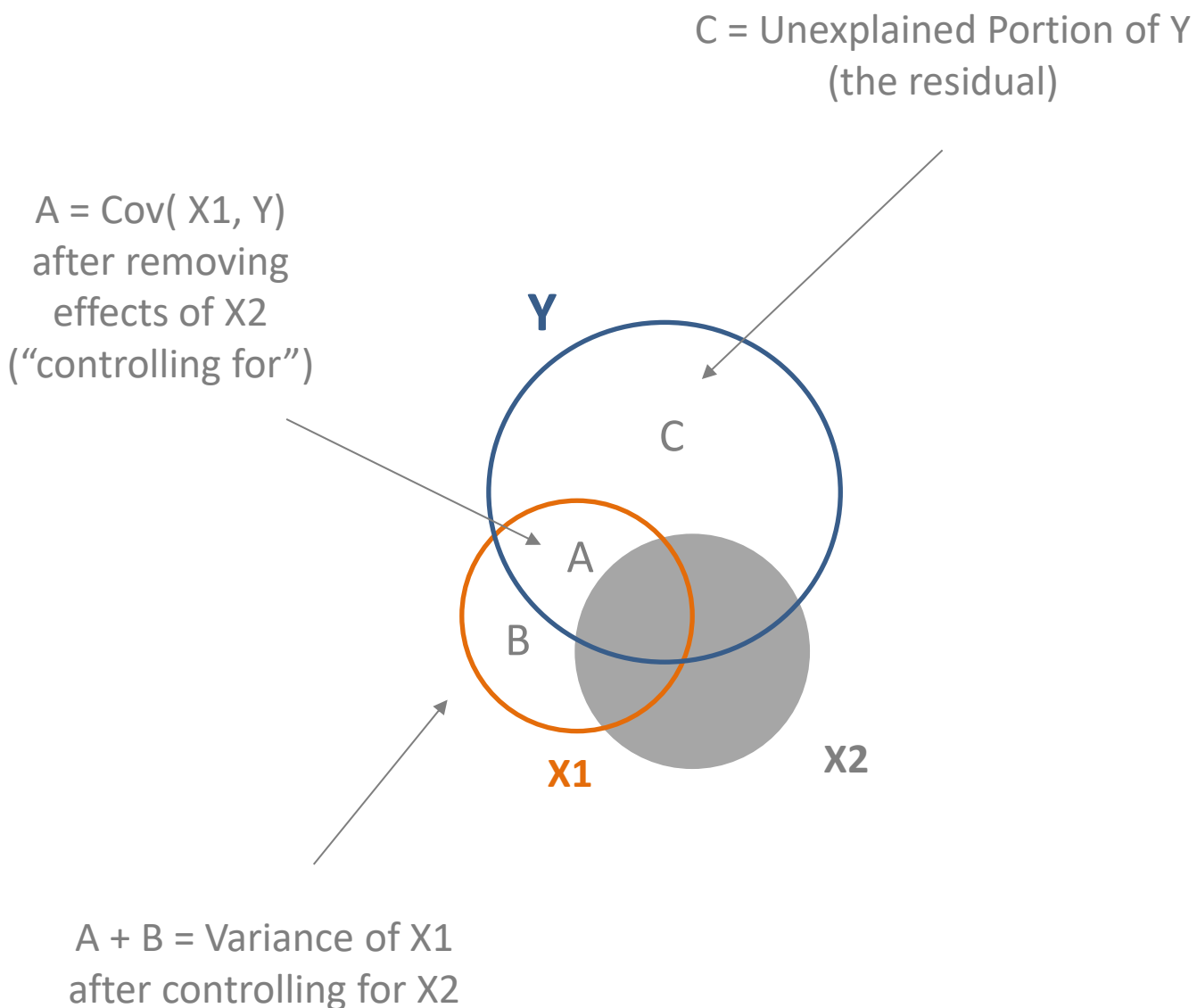
Relationship Between
Class Size and Test Scores

Relationship After Controls

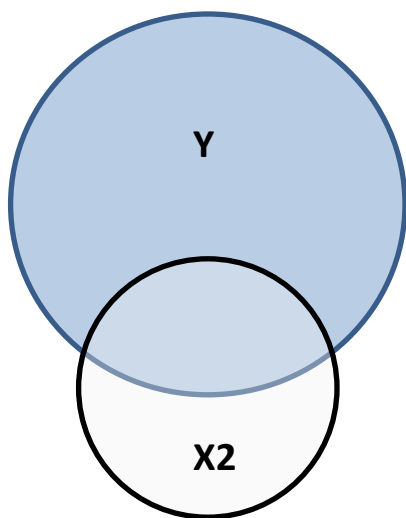




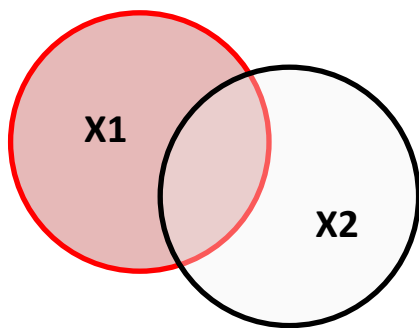
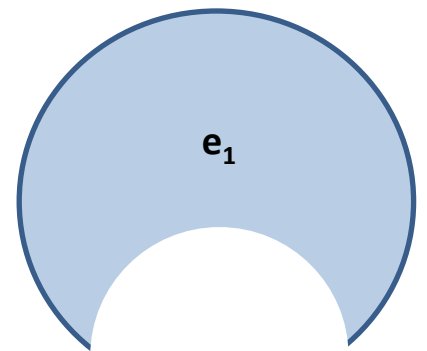
BALLENTINE VENN DIAGRAM



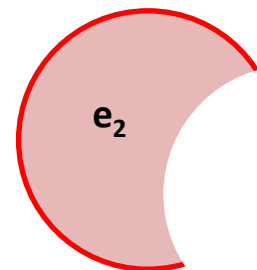
PARTITIONED REGRESSION



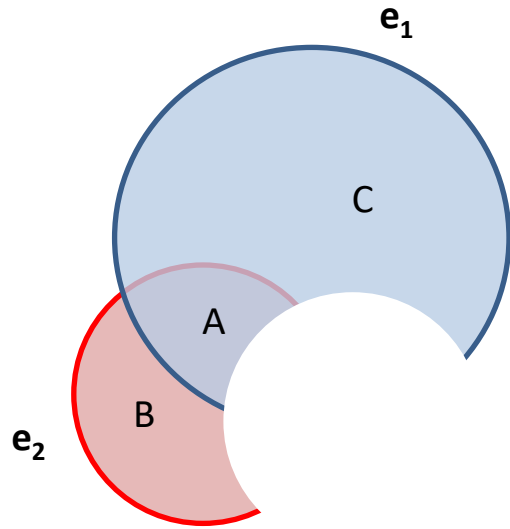
$$Y = a_0 + a_1 X_2 + e_1$$



$$X_1 = b_0 + b_1 X_2 + e_2$$



PARTITIONED REGRESSION



$$e_1 = \alpha_0 + \beta_1 e_2 + \varepsilon$$

$$\beta_1 = \frac{A}{A+B}$$

$$\varepsilon = C$$

WHY CAN WE DO THIS?

X	Y	$\hat{Y} = 0.6 + 0.9x$	e	e^2
2	3	$0.6 + 0.9 \cdot 2 = 2.4$	.6	.36
4	3	4.2	-1.2	1.44
6	6	6	0	0
8	9	7.8	1.2	1.44
10	9	9.6	-.6	.36
Mean=6	Mean=6	Mean=6	Sum=0	Total= 3.6

The residual is just another variable.

The order of operations is important in partitioned regression.

NOTE: MULTIPLE REGRESSION CONTEXT

$$Y = a_0 + a_1X_2...+ a_kX_k + e_1$$

$$X_1 = b_0 + b_1X_2...+ b_kX_k + e_2$$

If there is more than one control variable
make sure you cleanse all of the nuisance
variance from both your dependent variable
and your policy variable.

EXAMPLE 1:

$$SWB = \beta_0 + \beta_1 \text{PSS} + \beta_2 ND + \beta_3 NP + \varepsilon$$

“policy variable”

Coefficients ^a						
Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	-1.507	2.633		-.572	.567
	Perceived Social Support	.310	.032	.460	9.645	.000
	Network Diversity	-.215	.210	-.055	-1.025	.306
	# People in Social Network	.023	.014	.087	1.654	.099

Dependent Variable: **Subjective Well Being**

AUXILIARY REGRESSION

#1

$$SWB = b_0 + b_1ND + b_2NP + e_1$$

		Coefficients ^a				
		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
Model		B	Std. Error	Beta		
1	(Constant)	21.630	1.207		17.915	.000
	Network Diversity	.224	.228	.057	.979	.328
	# People in Social Network	.039	.016	.148	2.540	.011

Dependent Variable: **Subjective Well Being**

Regress the DV on all IV's except the policy variable.
Save the residuals.

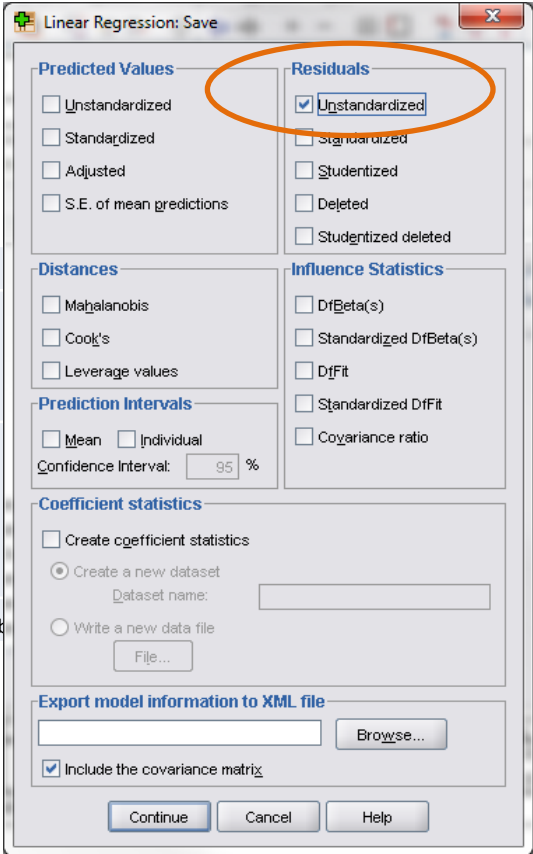
AUXILIARY REGRESSION

#1

Model

1

a. Dependent Variable



The image shows the 'Linear Regression: Save' dialog box in SPSS. The 'Residuals' section is circled in orange, and the 'Unstandardized' checkbox is checked. Other options in the 'Residuals' section include Standardized, Studentized, Deleted, and Studentized deleted. The 'Predicted Values' section has checkboxes for Unstandardized, Standardized, Adjusted, and S.E. of mean predictions. The 'Distances' section has checkboxes for Mahalanobis, Cook's, and Leverage values. The 'Prediction Intervals' section has checkboxes for Mean and Individual, with a confidence interval of 95%. The 'Coefficient statistics' section has a checkbox for 'Create coefficient statistics' and radio buttons for 'Create a new dataset' (selected) and 'Write a new data file'. The 'Export model information to XML file' section has a text box and a 'Browse...' button, with a checked checkbox for 'Include the covariance matrix'. At the bottom are 'Continue', 'Cancel', and 'Help' buttons.

+ e_1

Standardized Coefficients			
Beta	t	Sig.	
	17.915	.000	
.057	.979	.328	
.148	2.540	.011	



AUXILIARY REGRESSION

#2

$$PSS = a_0 + a_1ND + a_2NP + e_2$$

Coefficients ^a						
Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	74.720	1.720		43.438	.000
	Network Diversity	1.417	.325	.244	4.356	.000
	# People in Social Network	.052	.022	.133	2.364	.019
Dependent Variable: Perceived Social Support						

Regress the policy variable on all of the IV's.
Save the residuals.

PARTITIONED REGRESSION

$$e_1 = c_0 + \beta_1 e_2 + \varepsilon$$

Regress the residuals on each other.

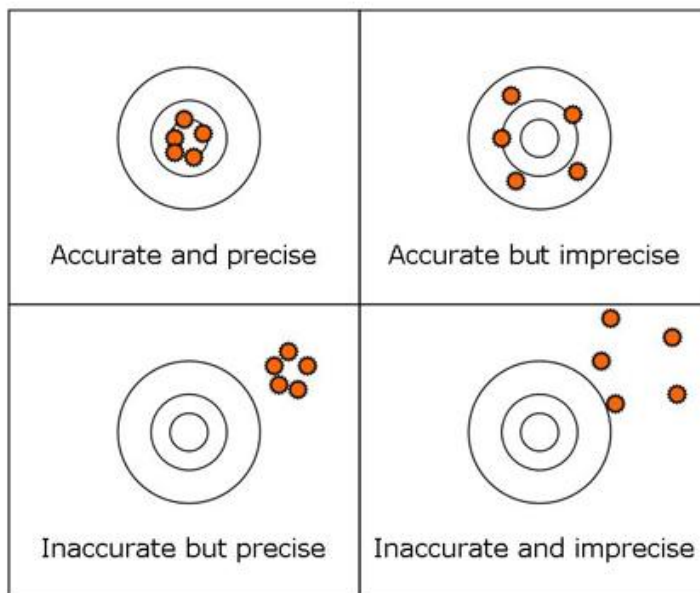
		Coefficients ^a				
		Unstandardized Coefficients		Standardized Coefficients		
Model		B	Std. Error	Beta	t	Sig.
1	(Constant)	2.369E-15	.253		.000	1.000
	Unstandardized Residual	.310	.032	.441	9.670	.000
Dependent Variable: Unstandardized Residual						

We have recovered the original slope!

WHY IS THIS IMPORTANT?

- It illustrates the idea of statistical controls in policy work – get rid of the nuisance variance!
- Provides insight into the regression error term.
- Serves as a tool for future methods (instrumental variables).
- It paved the way for modern computational econometrics.

WHY WE CARE:



Results need to be:

- Unbiased (accurate)
- Efficient (precise)

Biased and inefficient

Model 1

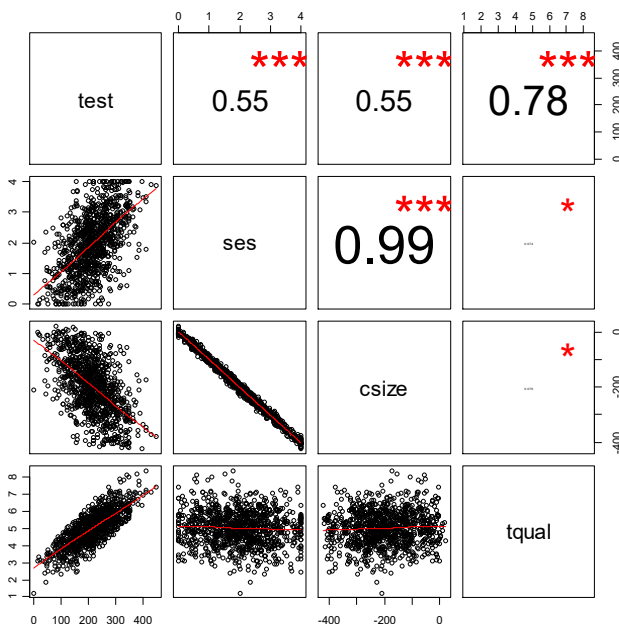
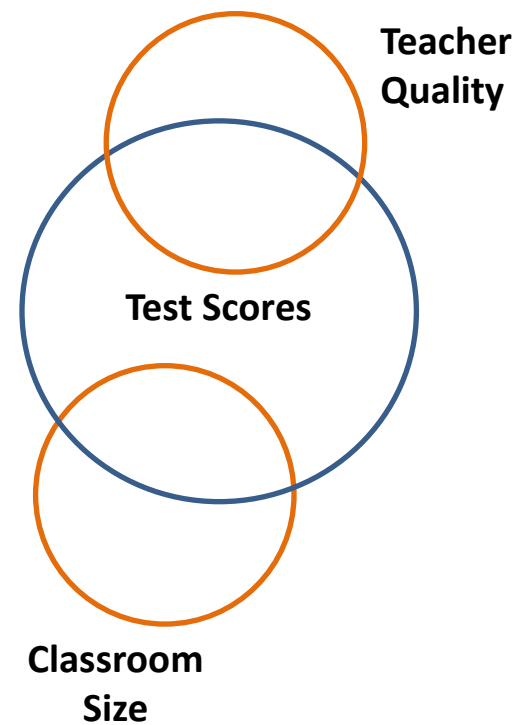
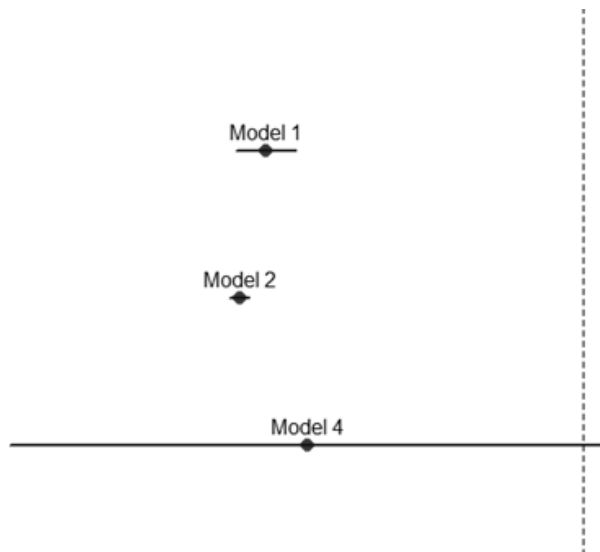
Model 2

Biased and efficient

Model 4

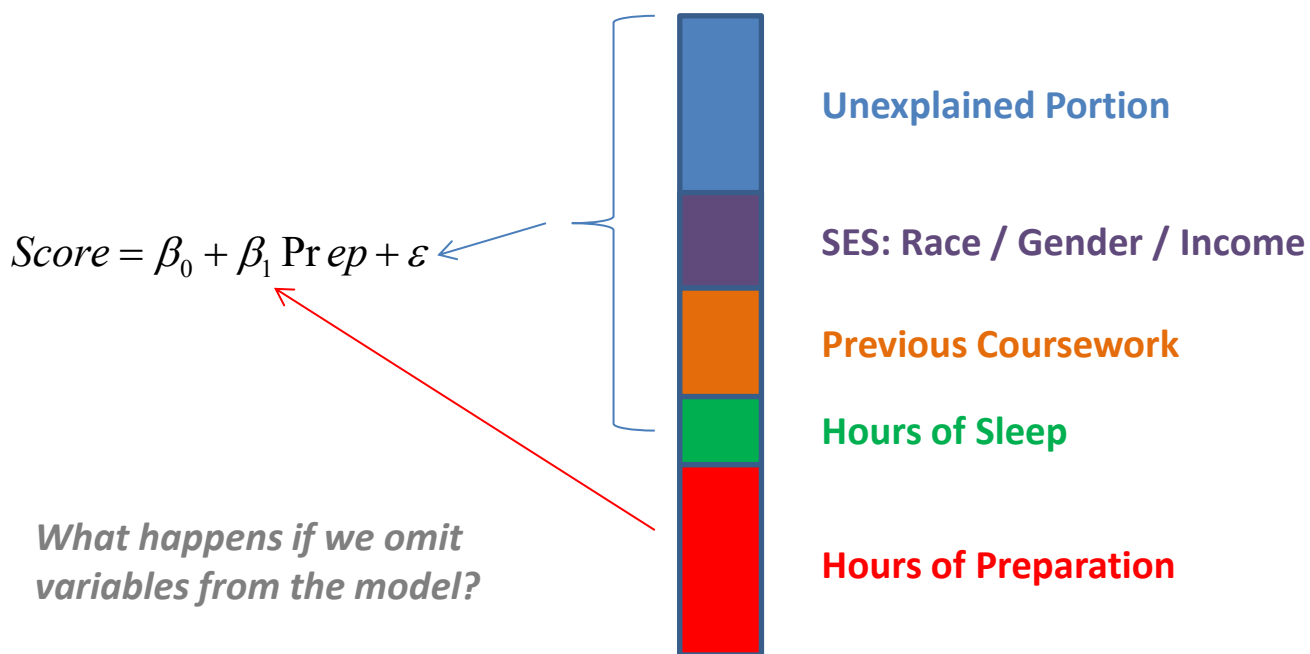
Unbiased and inefficient

EFFECTS OF CONTROLS DETERMINED BY THEIR CORRELATIONS WITH THE OUTCOME AND POLICY VARIABLES

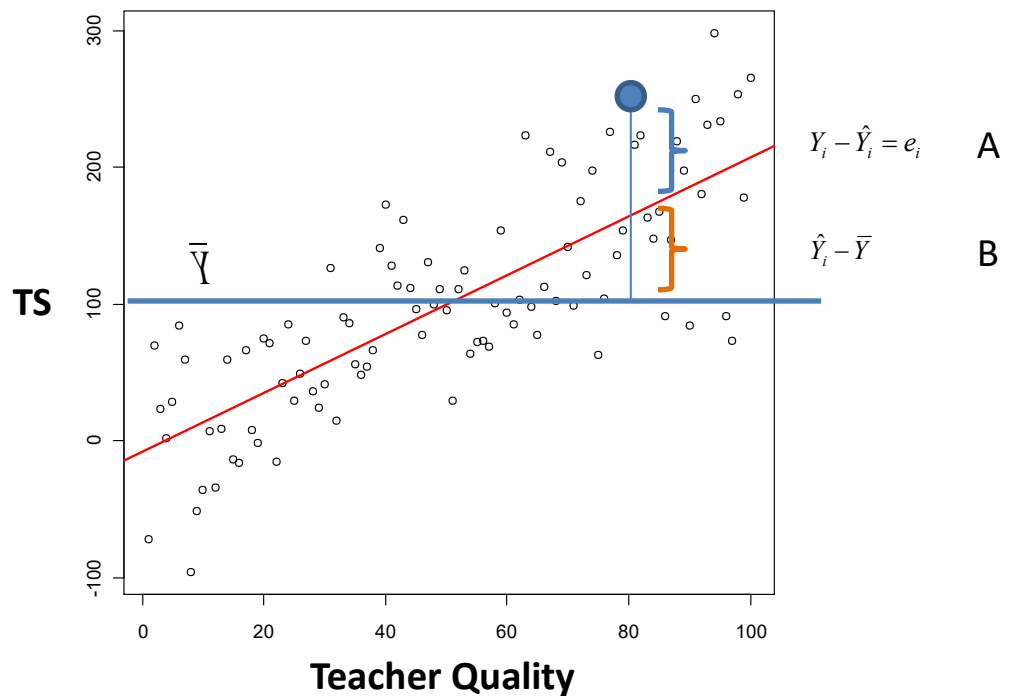
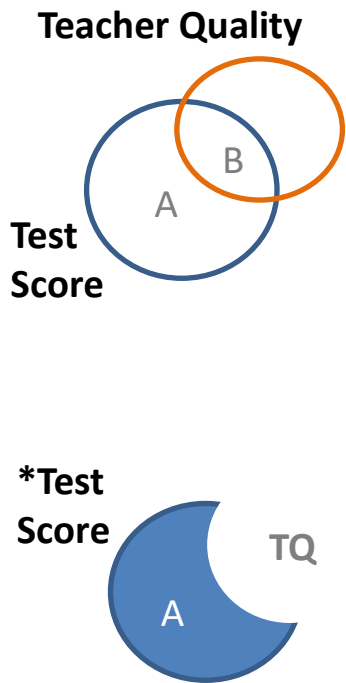


PARTITIONING THE VARIANCE OF Y

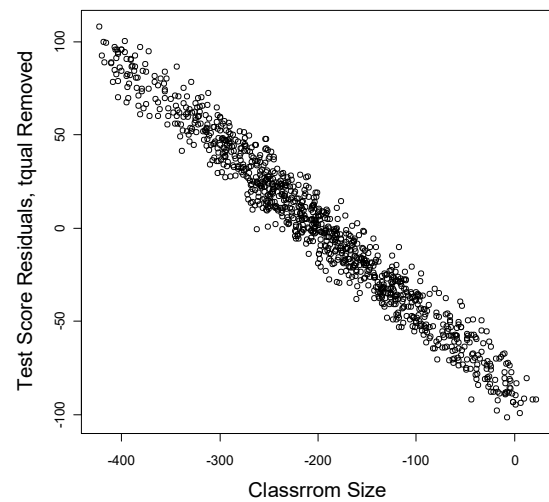
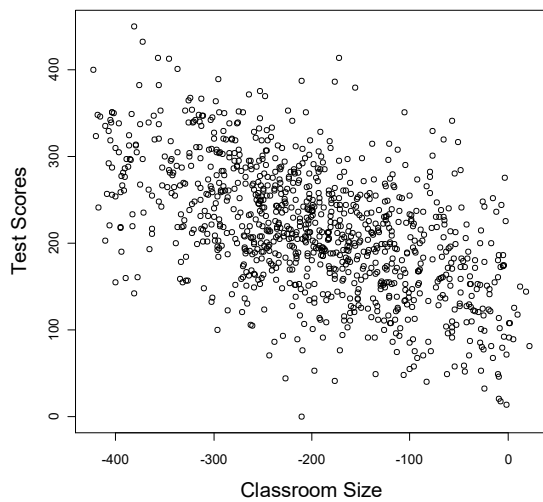
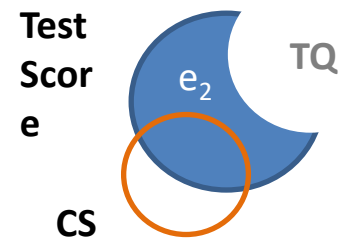
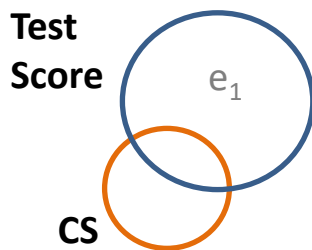
Final Exam Scores



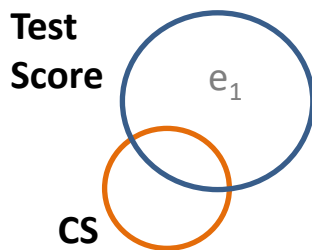
PARTITIONING THE VARIANCE OF Y



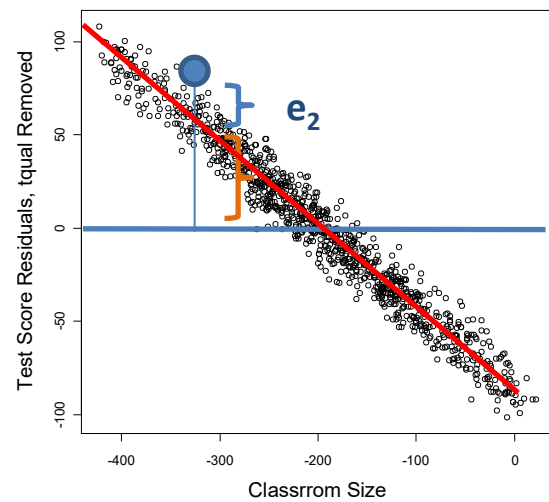
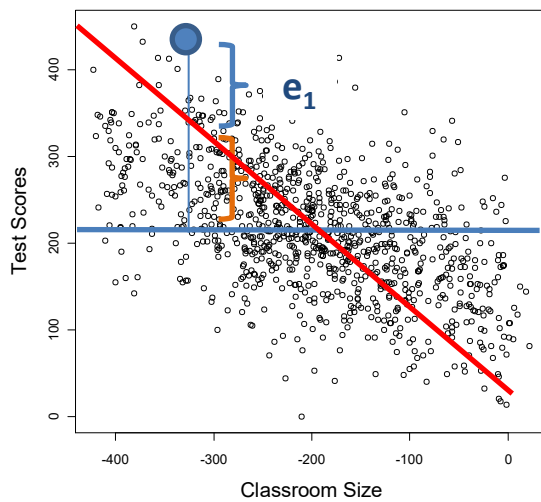
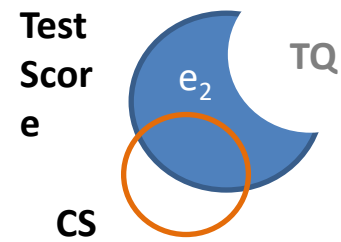
EFFECTS OF THE CONTROL VARIABLE



EFFECTS OF THE CONTROL VARIABLE

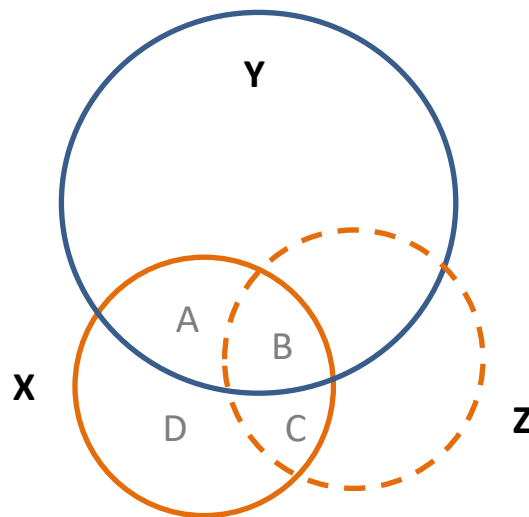


$$e_1 > e_2$$



PARTITIONING Y WITH MULTIPLE VARIABLES

WHAT HAPPENS TO SLOPES AND STANDARD ERRORS WHEN WE ADD MORE VARIABLES?

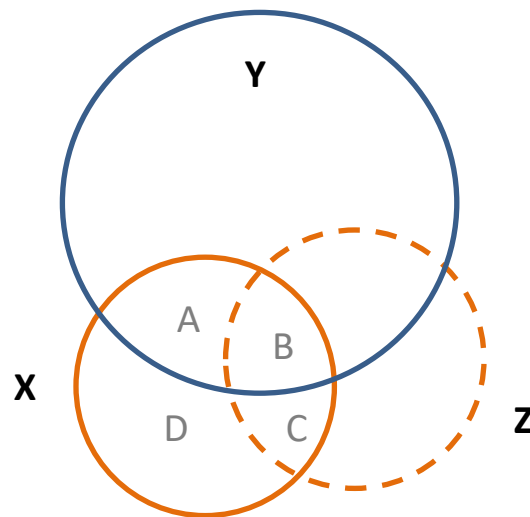


MULTIPLE REGRESSION MECHANICS: SLOPE

$$Y = b_0 + b_1X + e$$

$$b_1 = \frac{\text{cov}(x, y)}{\text{var}(x)}$$

$$b_1 = \frac{A + B}{A + B + C + D}$$



$$Y = \beta_0 + \beta_1X + \beta_2Z + \varepsilon$$

$$\beta_1 = \frac{r_{yx} - r_{yx}r_{xz}}{1 - r_{xz}^2}$$

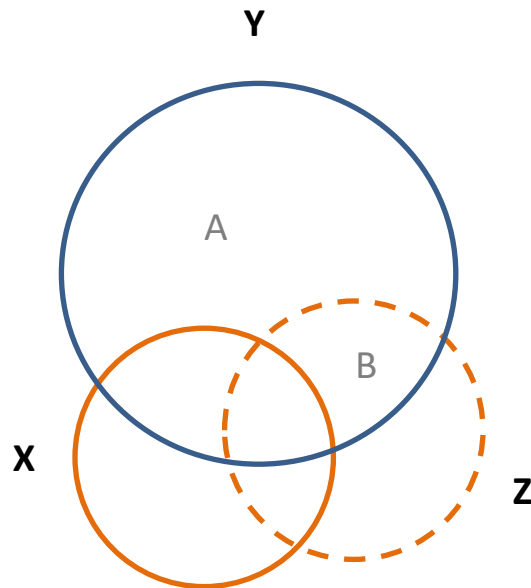
$$\beta_1 = \frac{A}{A + D}$$

The formula for the slope is different when you add more variables

MULTIPLE REGRESSION MECHANICS: STANDARD ERRORS

$$Y = b_0 + b_1X + e$$

$$e = A + B$$



$$Y = \beta_0 + \beta_1X + \beta_2Z + \varepsilon$$

$$\varepsilon = A$$

The unexplained variance is different when you add more variables



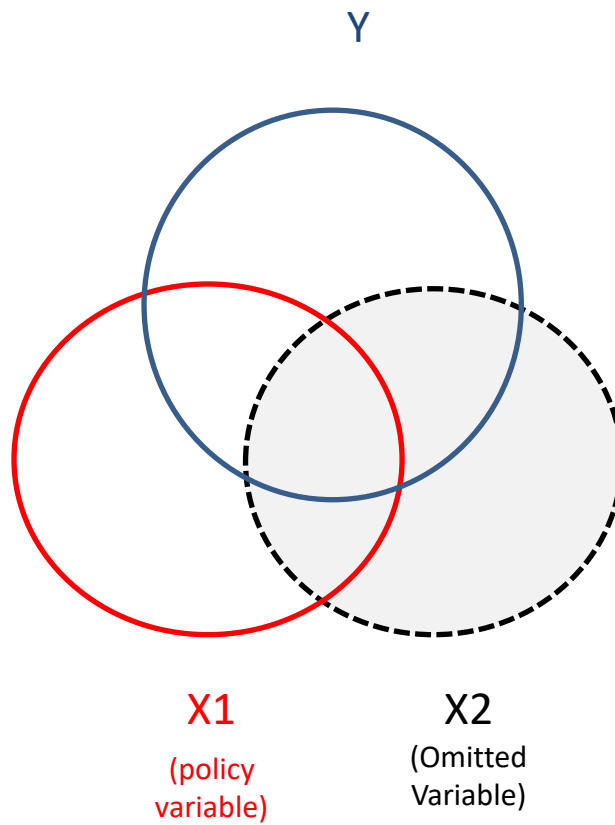
BIAS CORRECTION THROUGH INSTRUMENTAL VARIABLES

Fundamentals of
PROGRAM EVALUATION

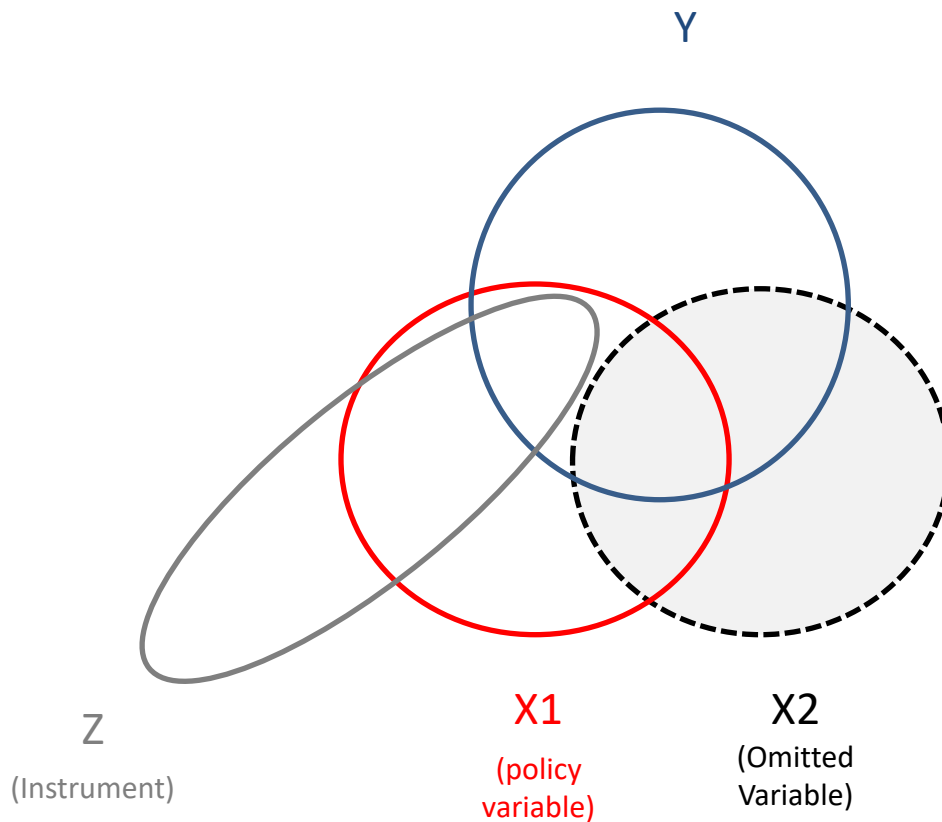
JESSE LECY

THREE MINUTE VERSION:

WE HAVE AN OVB PROBLEM:



WE HAVE AN INSTRUMENTAL VARIABLE:

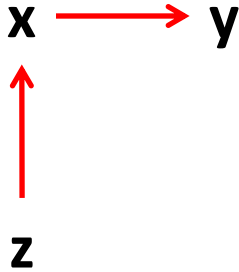


Z is correlated with X1 (the policy variable),
uncorrelated with X2 (the omitted variable).

A Venn diagram illustrating the relationship between four variables: Z, Y, X1, and X2. The diagram consists of four overlapping circles. Circle Z is gray and labeled 'Z (Instrument)' at the bottom left. Circle Y is blue and labeled 'Y' at the top. Circle X1 is red and labeled 'X1 (policy variable)' at the bottom center. Circle X2 is black with a dashed outline and labeled 'X2 (Omitted Variable)' at the bottom right. The circles overlap in various combinations, with the intersection of X1 and X2 shaded in light gray.

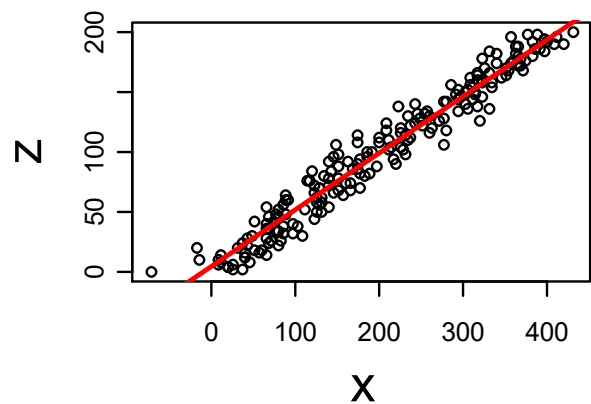
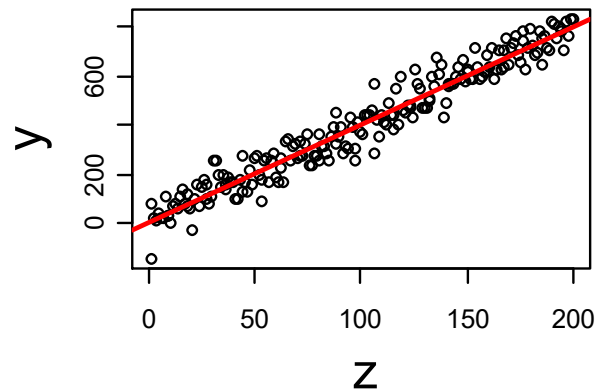
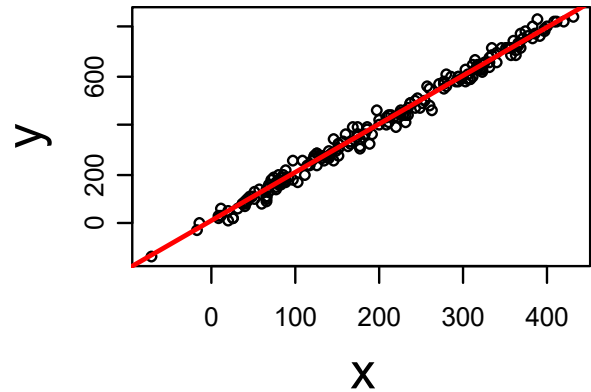
There is no causal relationship between Z and Y.

INSTRUMENTAL VARIABLES



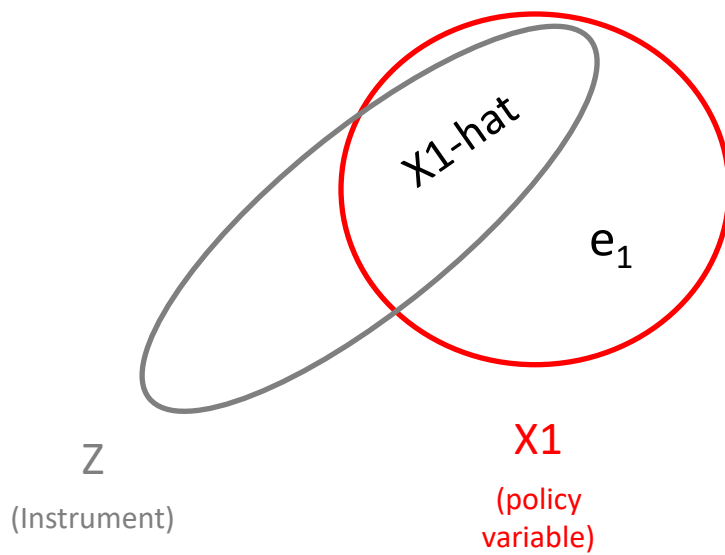
Correlation between Y and Z is natural extension of this causal structure.

In this data, Y has no direct causal relationship with Z, but there will be a strong correlation in the data because of Y and Z are both correlated with X.



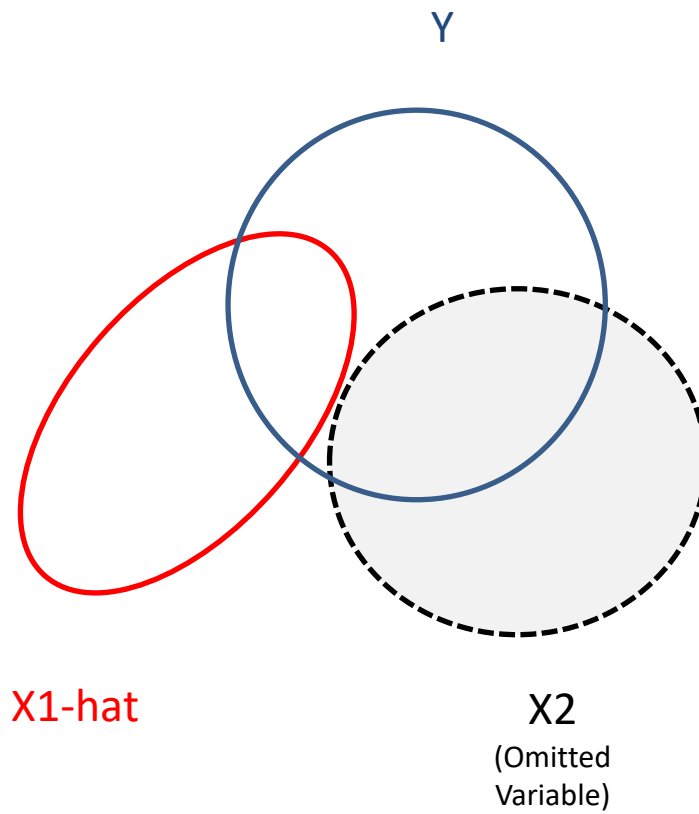
STEP 1:

Partition X_1 into parts correlated and uncorrelated with Z



$$X_1 = a_0 + a_1 * Z + e_1$$

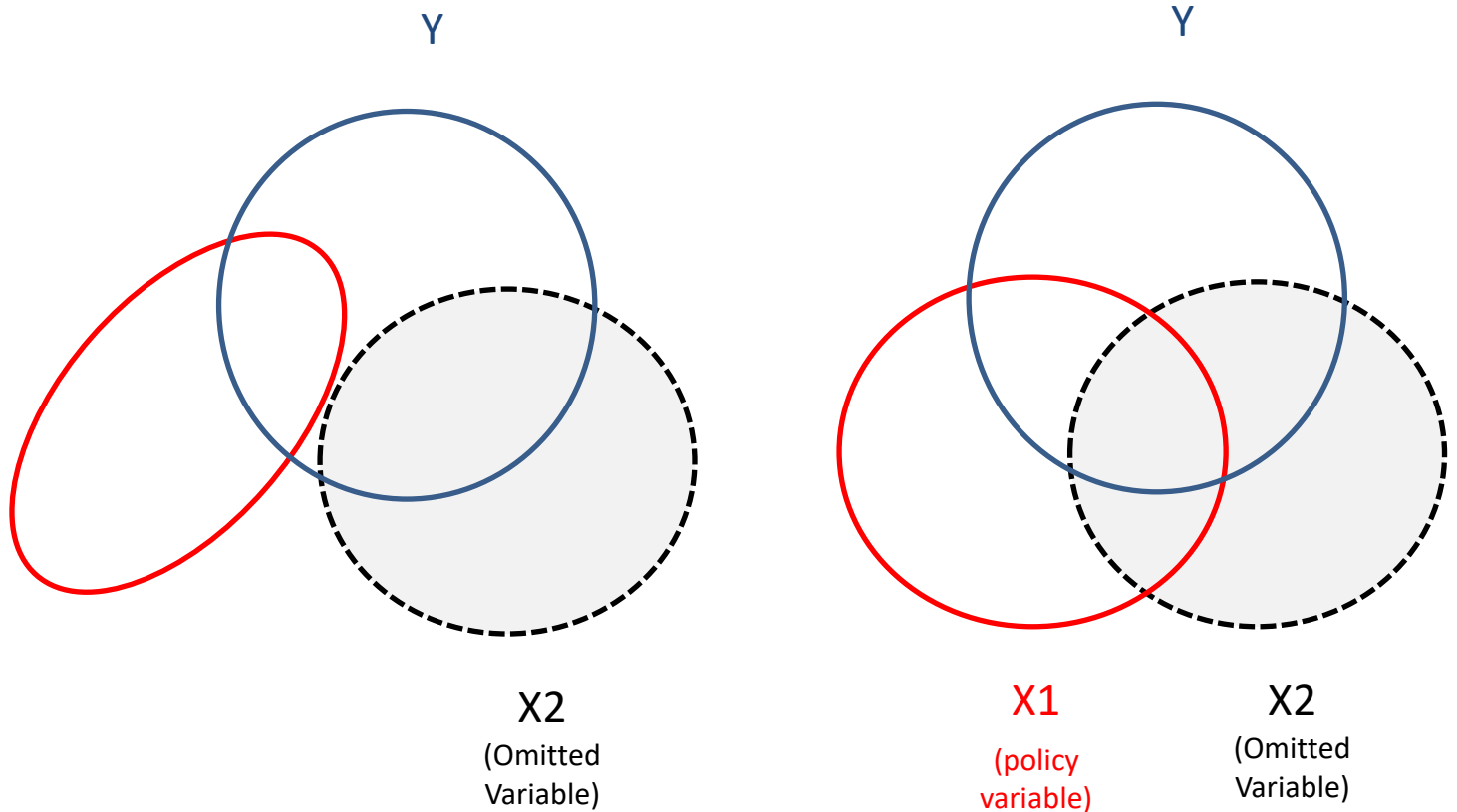
STEP 2:



$$Y = b_0 + b_1 * X1\text{-hat} + e_2$$

Keep only the part of X1 that is correlated with Z.

OVB MITIGATED:



X1-hat

$$Y = B_0 + B_1 * X1 + B_2 * X2 + e$$

Full Model

$$Y = b_0 + b_1 * X1\text{-hat} + e_2$$

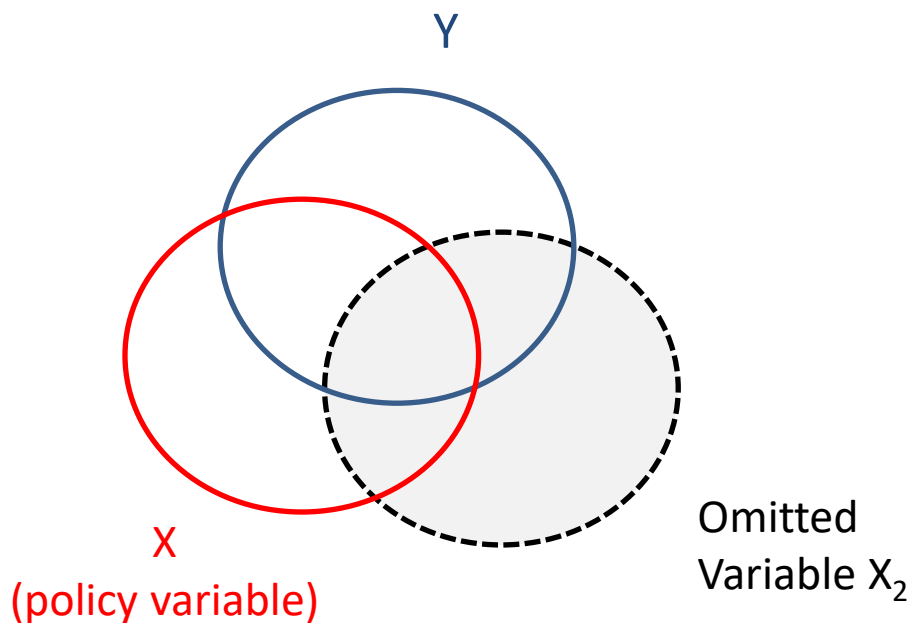
IV Model

$$b_1 \approx B_1$$

Naïve slope close to the true slope

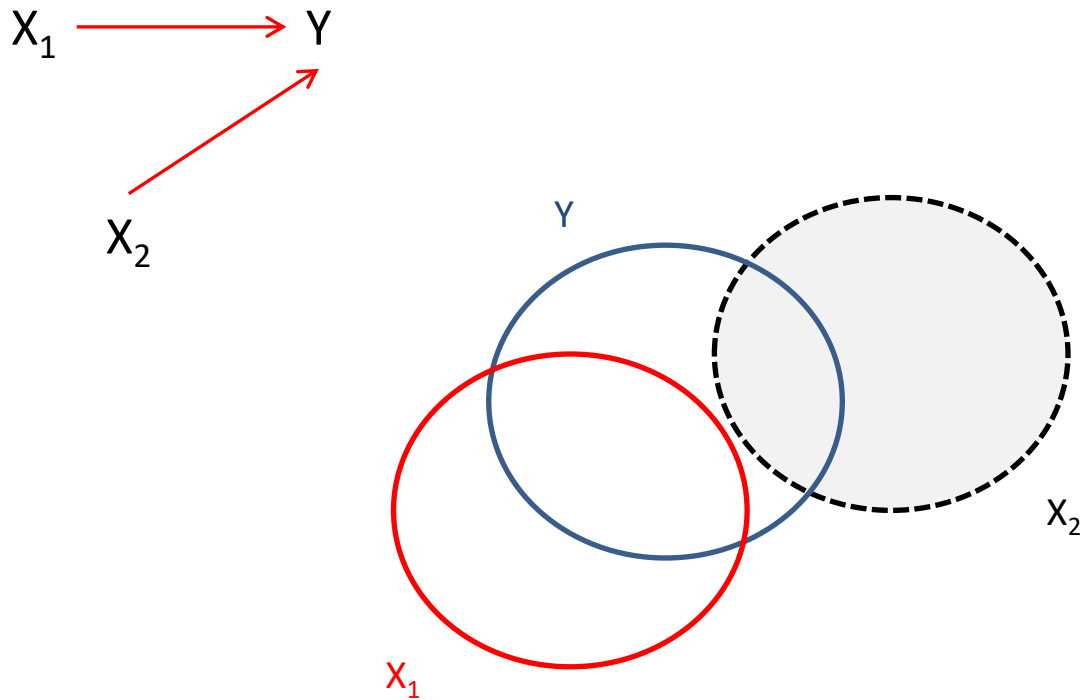
FULL VERSION:

WE HAVE USED A CORRELATION MODEL TO EXAMINE OMITTED VARIABLE BIAS

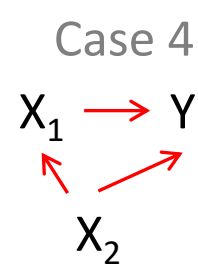
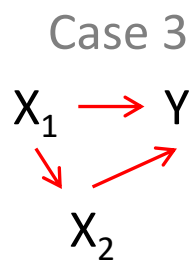
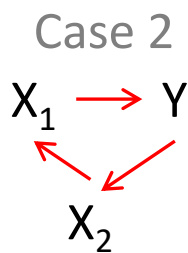


BUT THERE ARE MANY POSSIBLE CAUSAL MODELS:

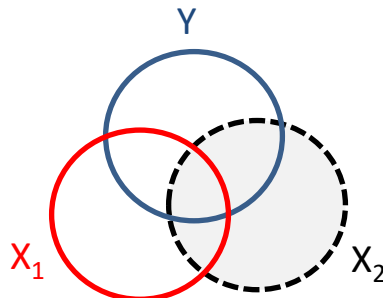
Case 1: Not Problematic



BUT THERE ARE MANY POSSIBLE CAUSAL MODELS:

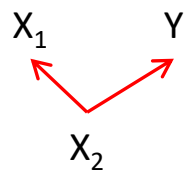


These are problematic when we omit X_2
because our slope on X_1 will not be biased

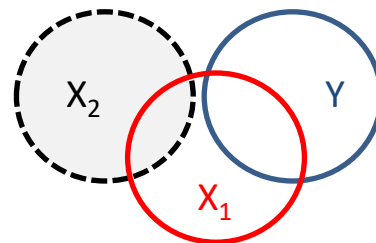
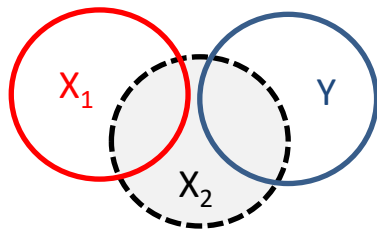
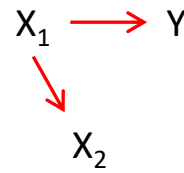


LURKING VARIABLES...SPURIOUS CORRELATIONS

Case 5



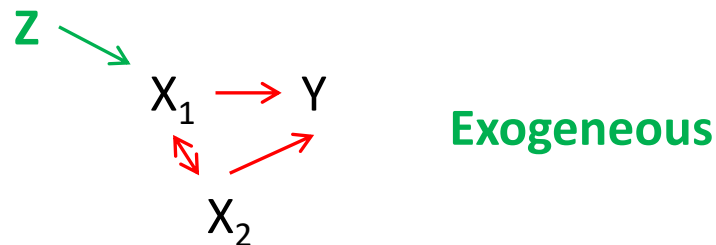
Case 6



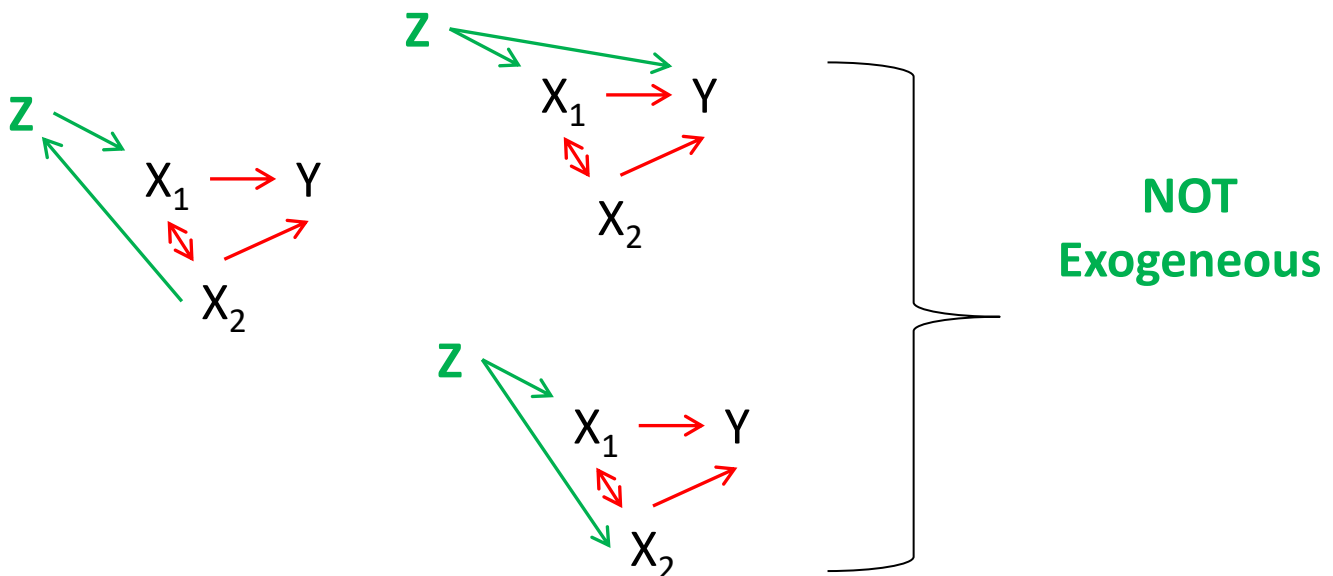
Don't worry about these cases for now

SO HOW DO WE FIX THIS?

INSTRUMENTAL VARIABLES EXPLOIT EXOGENEITY:



Exogenous: Variable that is correlated with the policy variable but NOT with the omitted variable.

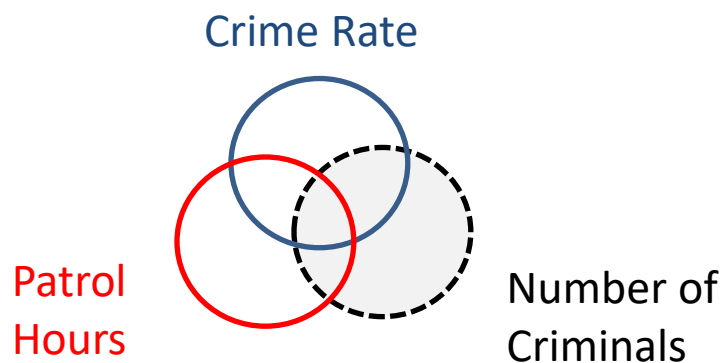


EXAMPLE:

Where have all the criminals gone?

Steven Levitte makes the claim that increasing policing only reduces crime slightly.

Why will patrol hours and the number of criminals be correlated? Will there be a positive or negative relationship?

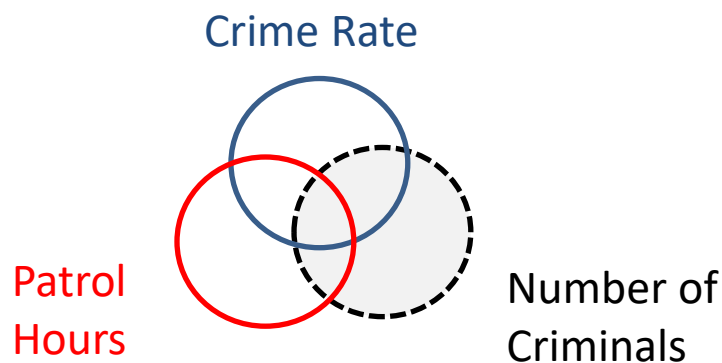


EXAMPLE:

Where have all the criminals gone?

Steven Levitte makes the claim that increasing policing only reduces crime slightly. The problem with the estimation:

“Cities with high crime rates, therefore, may tend to have large police forces, even if police reduce crime. Detroit has twice as many police officers per capita as Omaha, and a violent crime rate over four times as high, but it would be a mistake to attribute the differences in crime rates to the presence of the police. Similarly, within a particular city, if more police are hired when crime is increasing, a positive correlation between police and crime can emerge, even if police reduce crime.”

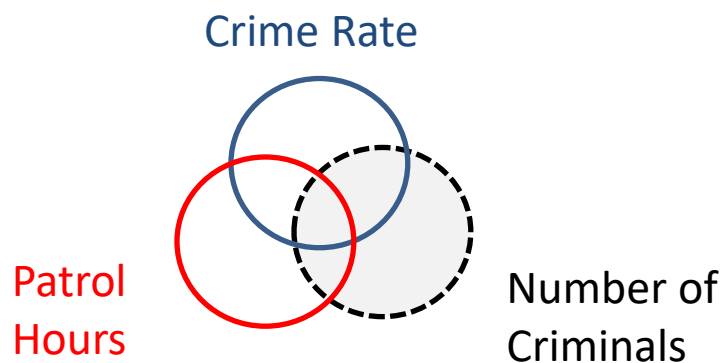


EXAMPLE:

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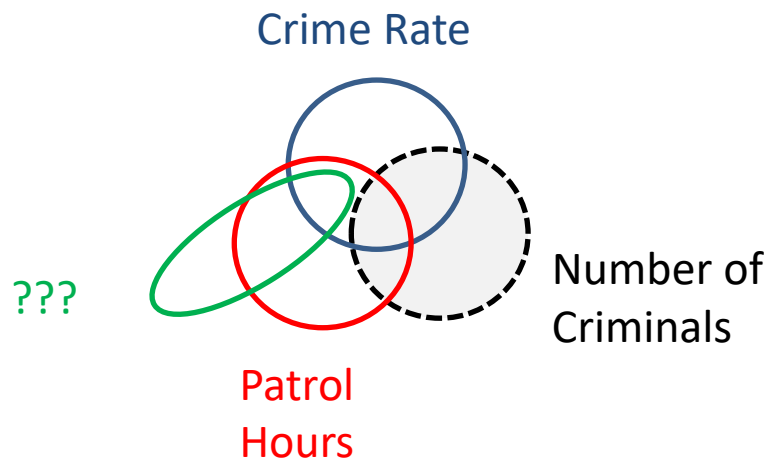
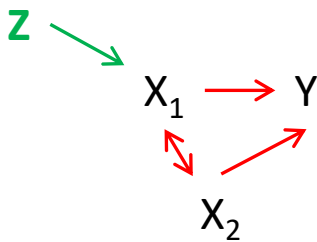
Where would we find a measure of the number of criminals?



THE FIX:

Where have all the criminals gone?

We need to find an exogenous variable correlated to policing intensity but uncorrelated with crime rates

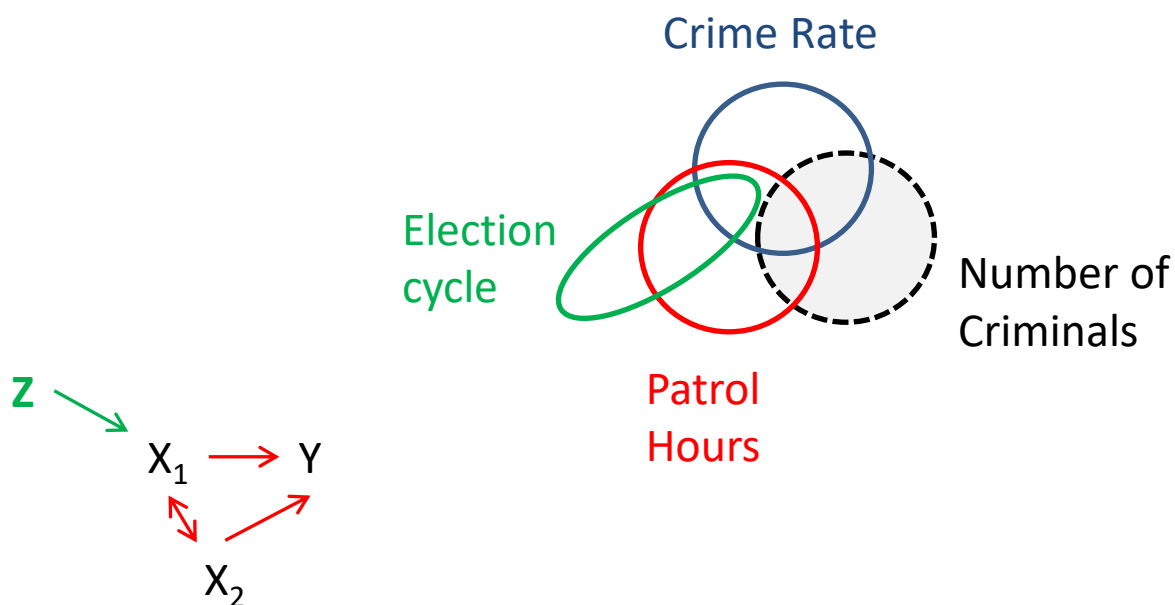


THE FIX:

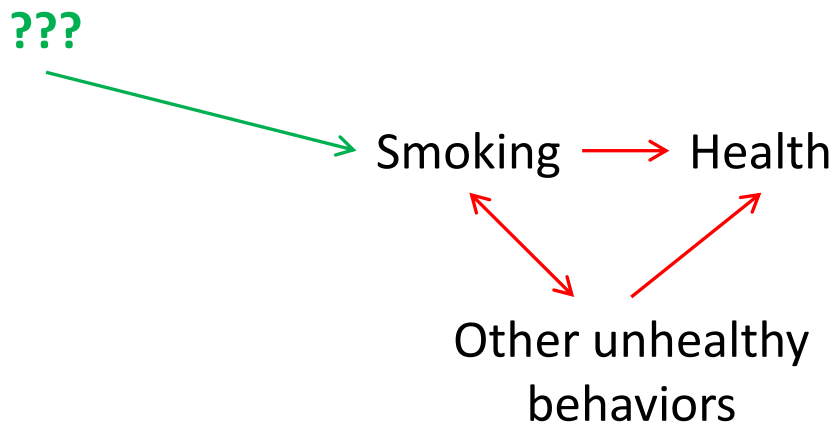
Where have all the criminals gone?

“The primary innovation of the paper is the approach used to break the simultaneity between police and crime. In order to identify the effect of police on crime, a variable is required that affects the size of the police force, but does not belong directly in the crime “production function.” The instrument employed in this paper is the timing of mayoral and gubernatorial elections.”

Levitt, 1997

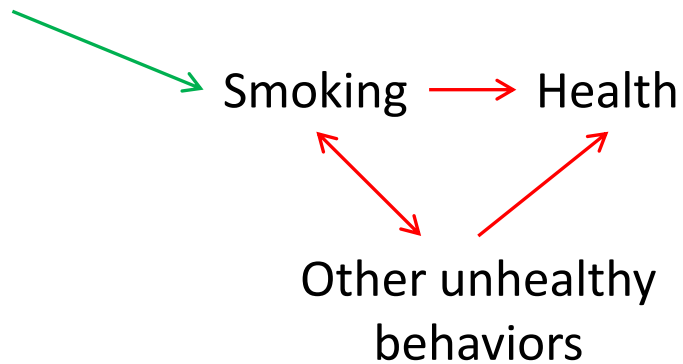


ANOTHER EXAMPLE:

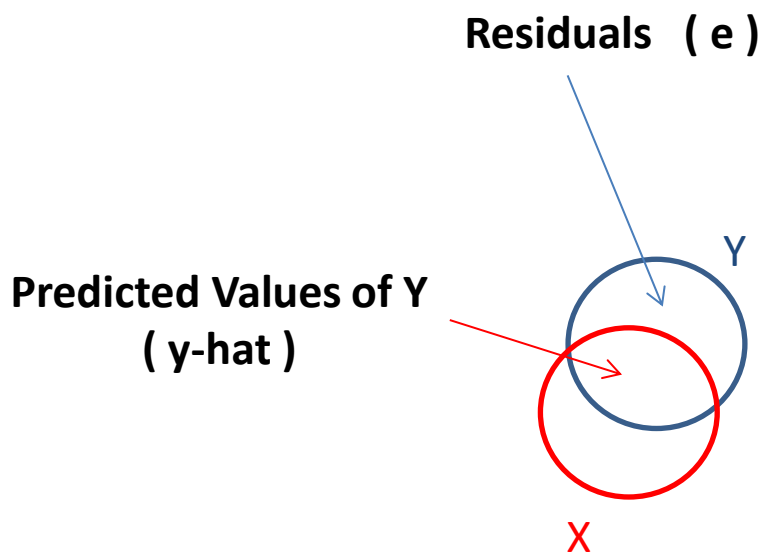


ANOTHER EXAMPLE:

Tobacco Taxes

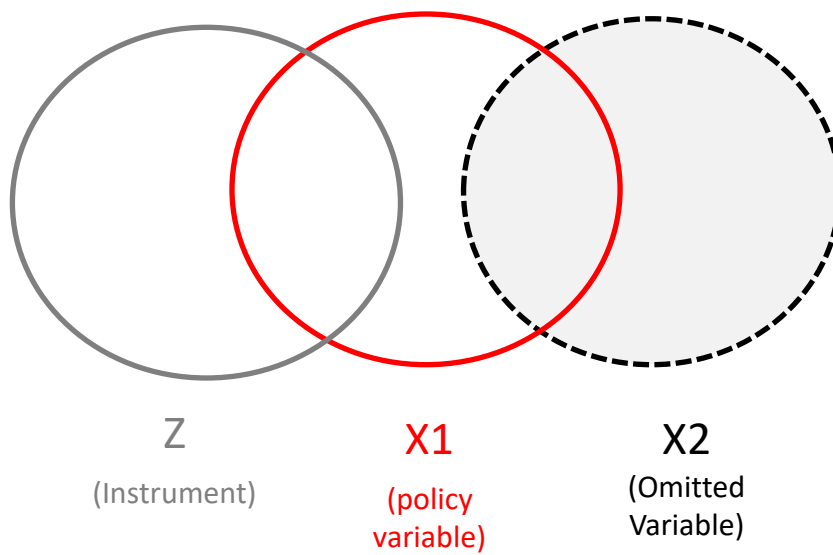


HOW DOES IT WORK?



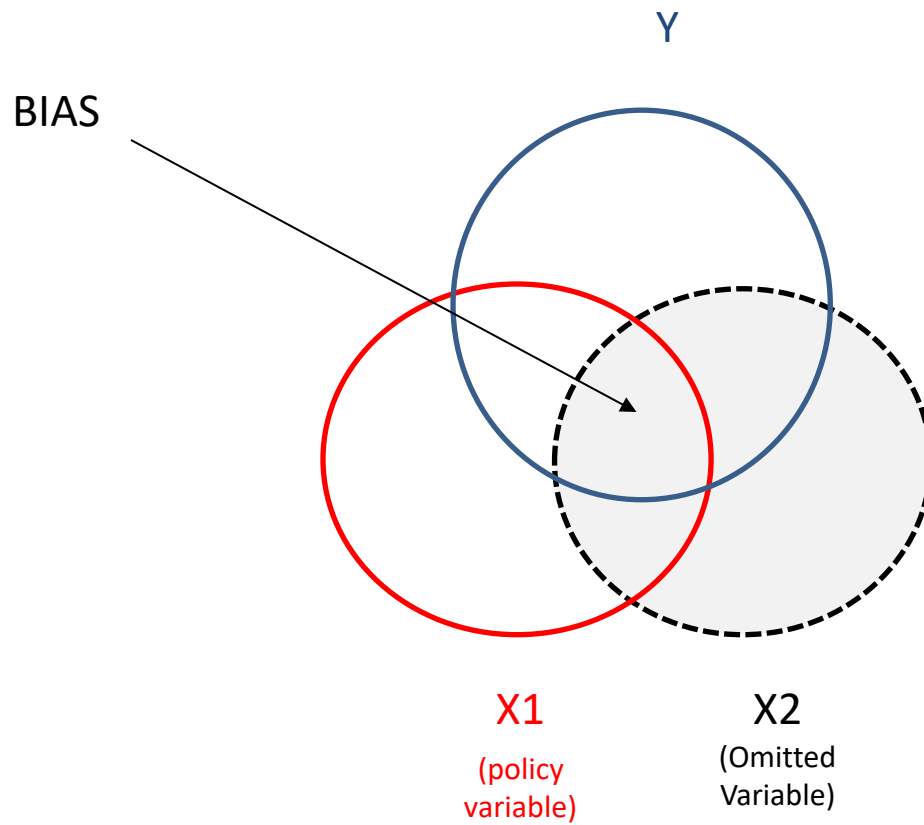
Recall, Y is partitioned into explained and unexplained portions

EXOGENOUS CRITERIA:



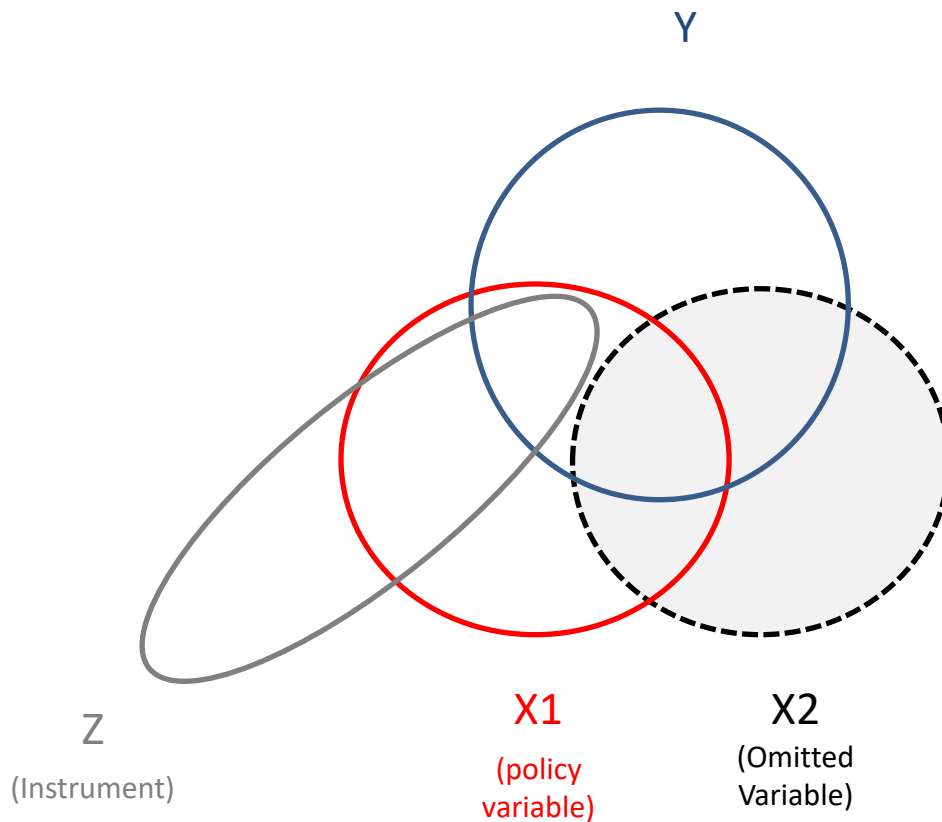
Z is uncorrelated with X2

EXOGENOUS CRITERIA:



X_2 is a problematic omitted variable

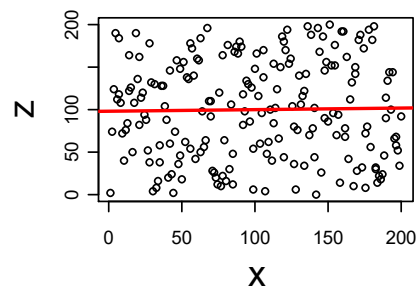
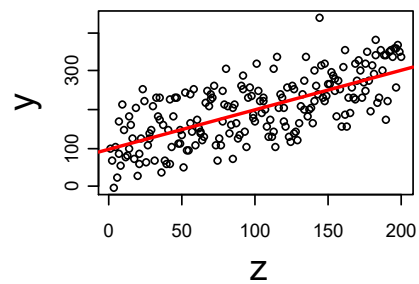
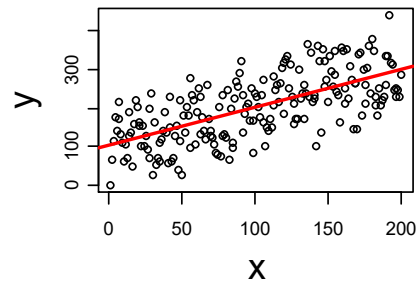
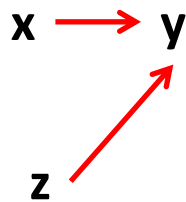
EXOGENOUS CRITERIA:



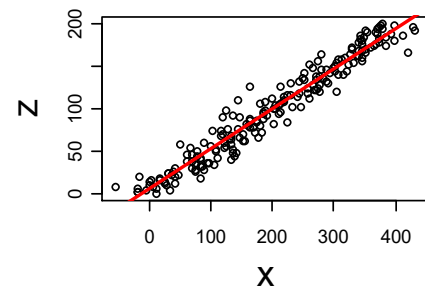
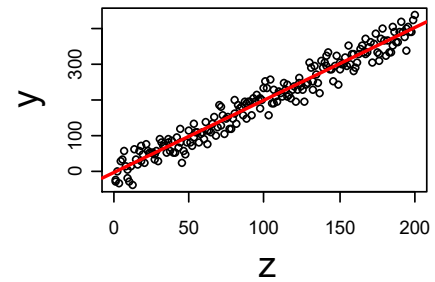
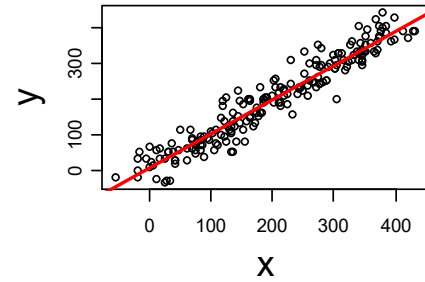
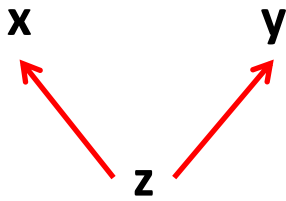
Z is correlated with Y, but only THROUGH X1

RECALL:

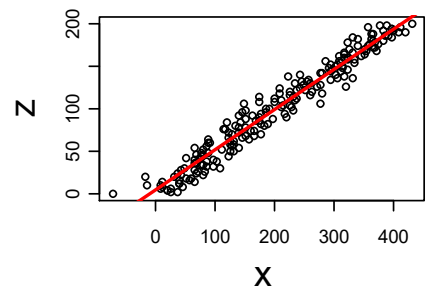
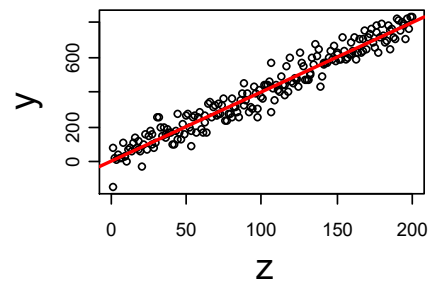
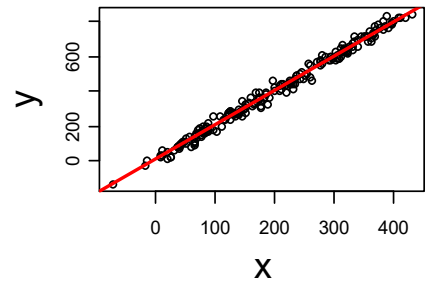
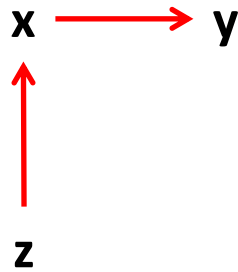
Example #1



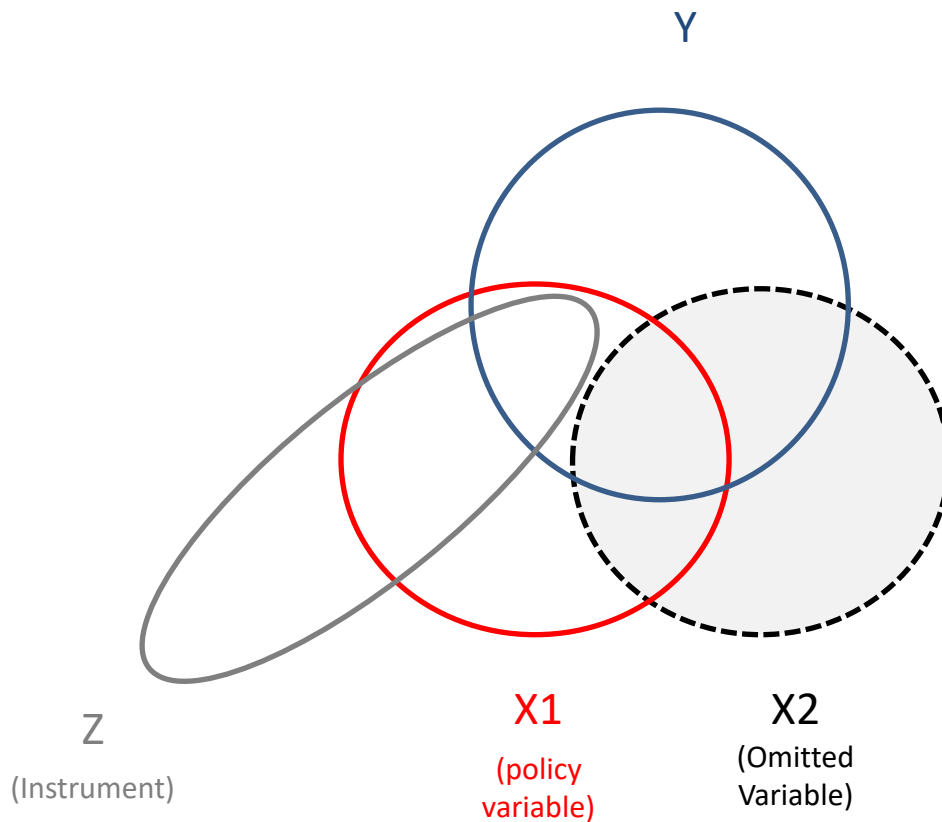
Example #2



Example #3



EXOGENOUS CRITERIA:

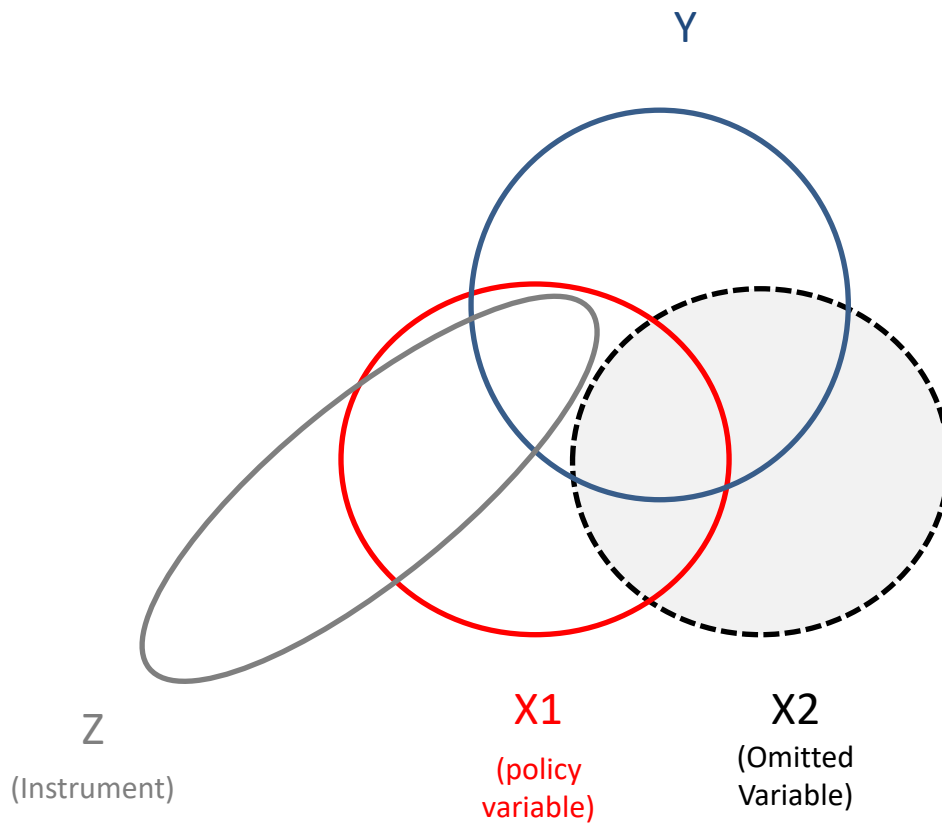


$$z \longrightarrow x \longrightarrow y$$

This is a natural property of correlation structures.

We exploit this property to isolate the uncontaminated portion of X1!

EXOGENOUS CRITERIA:



$$z \longrightarrow x \longrightarrow y$$

THIS IS HOW INSTRUMENTAL VARIABLES WORK:

True Model : $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \varepsilon$, $\text{Corr}(X_1, X_2) \neq 0$

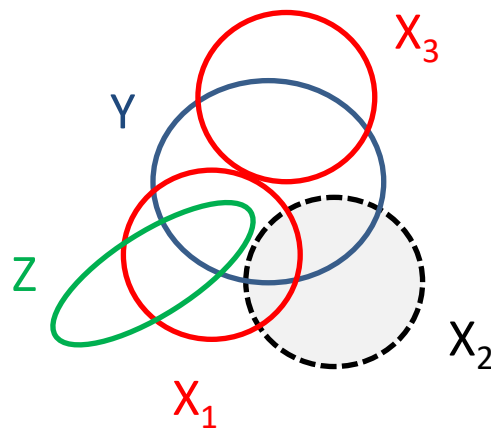
Biased Model : $Y = b_0 + b_1 X_1 + b_3 X_3 + e$

Two – Stage Least Squares

Stage 1 : $X_1 = \alpha_0 + \alpha_1 Z + \alpha_2 X_3 + e$

Stage 2 : $Y = \gamma_0 + \gamma_1 \hat{X}_1 + \gamma_3 X_3 + e$

Now : $\gamma_1 \approx \beta_1$



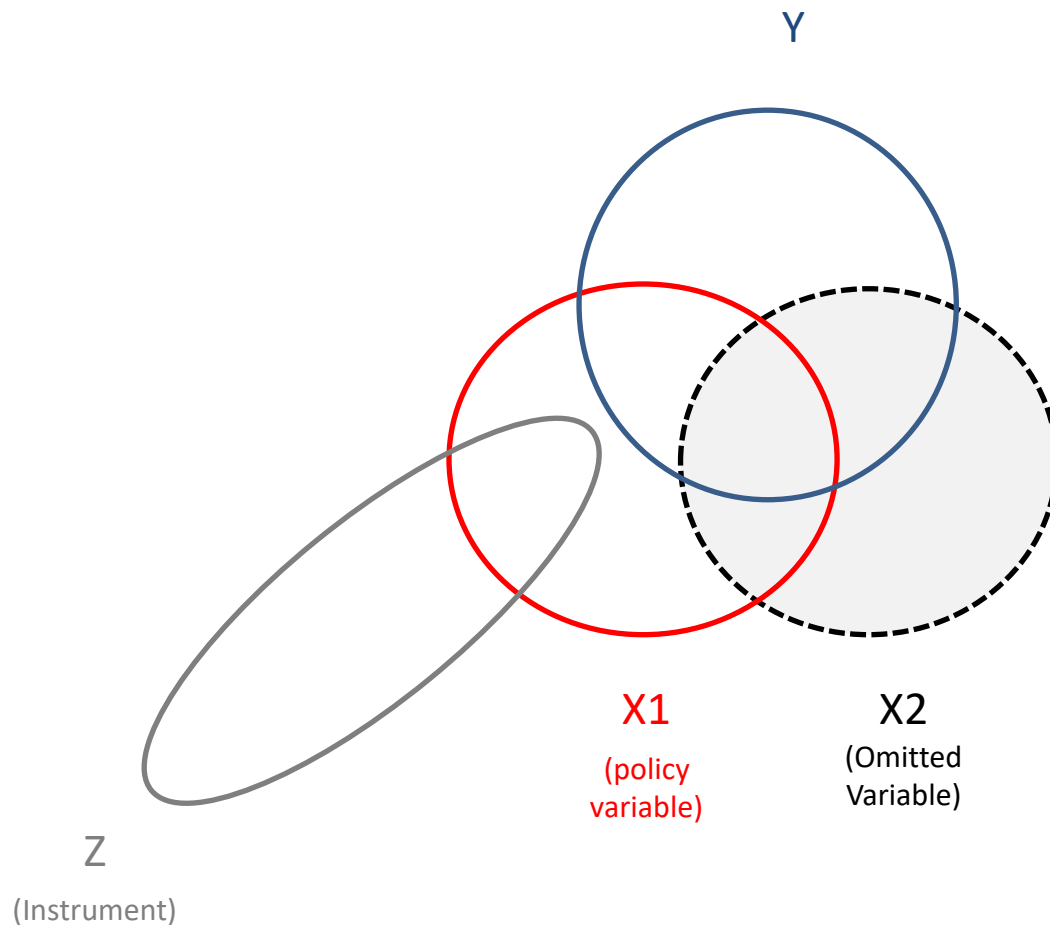
WE USE INSTRUMENTAL VARIABLES WHEN:

1. We know that we have an omitted variable problem
2. We don't have a way to include the omitted variable
3. We have an exogenous variable
 - But how do we know that we have an exogenous variable???
 - Instrumental variables are justified by assumption, i.e. a good story. There is a test you can run if you have multiple instrumental variables, but often we rely on the story.

PROCEDURE FOR IV APPROACH:

- 1) X_2 is the omitted variable – if it is omitted it means we don't have it so we can't include it in the model.
- 2) X_1 is the policy variable.
- 3) Z is the exogenous variable.
- 4) X_3, X_4, X_5 , etc. are all included in the model, but are not affected by the omitted variable in the same way that the policy variable is.
- 5) In the first stage, we predict model X_1 with the exogenous variable and the other variables.
- 6) In the second stage, we ONLY include predicted values of X_1 , because these predicted values will be independent of the influences of the omitted variable since Z and X_2 are uncorrelated.

EXOGENOUS CRITERIA:



This case would result from a weak correlation between Z and X_1 .

Z will not work as an instrument here.

THE THEORY:

- Using the instrument, we surgically remove the portion of X_1 that is “contaminated” by the omitted variable.
- Instrumental variables rely on the assumption of exogeneity, but this assumption can be tested statistically if there is more than one exogenous variable.
- The models needs at least on exogenous variable for every independent variable that is endogeneous.
- We need good predicted values for X_1 , so the R-square of the regression must be high (an F-stat of over 10 is the rule of thumb).